

Title

University of Glasgow



University
of Glasgow

Joseph Henry Briggs

Day Month Year

Contents

1	Characterisation of InGaAs Photodiodes for detecting 2μm Light	5
1.1	Introduction	5
1.2	What is an Extended InGaAs Photodiode	7
1.3	What is Needed from the Read-out Photodiode in a Gravitational Wave detector	10
1.4	Reverse Bias and the Quantum Efficiency of an Extended InGaAs Photodiode	13
1.4.1	Method	13
1.4.2	Results	16
1.5	Dark Current and Dark Noise in Extended InGaAs Photodiodes . . .	21
1.5.1	Room Temperature	22
1.5.2	Cooling Experiment	26
1.6	The Saturation Limit of Some Laser Components Extended InGaAs Photodiodes	27
1.7	Excess Noise in the IG24x500S4i	30
1.8	Properties of an InSb detector	35
1.9	Discussion	38

1.10 Conclusion	41
A Extended InGaAs Diodes Noise Spectrums	43
A.1 Bias Section	43
A.2 Temperature Section	48
A.3 InSb	48
B Photodiode Circuits	53
B.1 Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes	53
B.1.1 Transfer function	56
B.1.2 Noise measurement	58
B.2 Transimpedance Amplifier for Making Shot Noise limited Measure-	
ments of 10mA Photocurrents	58
B.3 Simulations with LISO	61
B.4 Transfer Function Calculation With MATLAB	61

Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Apparently this is a good chapter...
- Chapter ??: Passed first drafting. Very Minor corrections to be made.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and bryan comments
- Chapter ??: Not written, vaguely planned
- Chapter 1: Significant amounts of thought has already gone into this.
- Chapter 1.10: Not started. Will write last/almost last

Chapter 1

Characterisation of InGaAs

Photodiodes for detecting 2 μ m Light

1.1 Introduction

The current generation of gravitational wave detectors will be limited by the thermal noise due to the Brownian motion of the molecules that make up the mirror coatings. Since these thermally driven fluctuations are a limiting source of noise [1], proposed future gravitational wave detectors, such as the Einstein Telescope [2] and LIGO Cosmic Explorer [3], will use cryogenically cooled mirrors.

However, cryogenically cooled mirror coatings will not be compatible with the current laser wavelength, which is 1 μ m, and so future gravitational wave detectors may use either a 1.5 μ m or 2 μ m wavelength laser. 2 μ m is of interest as there have been promising designs for low noise mirror coatings that are optimised for this wavelength.

As InGaAs is a widely used semiconductor for detecting 1-1.6 μm light, suitable InGaAs photodiodes were available for detecting the signal-carrying light in the current generation of gravitational wave detectors. These photodiodes are low noise, linear, can operate at room temperature, can detect tens of milliwatts of light, and have $>99\%$ quantum efficiency.

As the current InGaAs photodiodes are not sensitive to 2 μm light, ‘extended’ InGaAs photodiodes may be used instead. These diodes are extended in the sense that their cut-off wavelengths are above 1.6 μm . To achieve this longer cut-off wavelength, extended InGaAs contains a higher percentage of indium compared to conventional InGaAs photodiodes; however, this extra indium comes at the expense of the structural integrity of the lattice. This can cause the quality of the photodiode to be degraded. Previous experiments have been limited by the quantum efficiency of the extended InGaAs photodiode that was used [4, 5]. To obtain low enough electronic noise in these experiments, the authors of [4] used the photodiodes without a bias – this came at the expense of the photodiodes quantum efficiency.

Alternative photodiode materials that could be used include HgCdTe and InSb. HgCdTe photodiodes with quantum efficiencies $>99\%$ are in the design stages of being made, however there are still manufacturing challenges to overcome. InSb detectors have high quantum efficiency [6], however this material would only be suitable for wavelengths above 2.6 μm .

In this chapter, the relationship between quantum efficiency, reverse bias and noise in off-the-shelf extended InGaAs photodiodes is explored. A selection of extended InGaAs photodiodes were tested with the aim of understanding general trends in the

behaviour of extended InGaAs photodiodes. It was found that to get a high quantum efficiency and linearity, the photodiodes must be operated with a bias. However, if the photodiodes were used with a high bias, their dark noise could exceed the shot noise level of the detected light. The dark noise could be reduced significantly by cooling the diodes and so future detectors may feature cooled read-out photodiodes. Future work may include investigating the growth conditions for extended InGaAs photodiodes as it has been shown that the dark noise of a diode made from extended InGaAs can be reduced by orders of magnitudes by optimising this. However, at the time of writing (2020) and to the authors knowledge, commercially available InGaAs photodiodes for detecting $2\text{ }\mu\text{m}$ light that are suitable for use as the read-out photodiode in a gravitational wave detector do not exist.

1.2 What is an Extended InGaAs Photodiode

A photodiode is a square law detector and so it produces a current that is proportional to the power of the light it is detecting. Usually, they are of the order 1 mm in diameter. They are two-terminal devices, with an anode and a cathode. Photodiodes often follow a p-doped, intrinsic, n-doped (PIN) structure. PIN photodiodes are used in applications where there is milliwatts of light power.

The basic physics of a PIN photodiode can be qualitatively understood by considering a simpler PN diode. A pn junction consists of two types of semiconductor: p-doped and n-doped. In the P-doped region, holes (the majority charge carriers) are free to move whereas in the n-doped region electrons (the majority charge carriers) are free to move.

At the interface between the p-doped and n-doped region, majority carriers from each side diffuse across it. Then, the minority charge carrier combines with a majority charge carrier, and so no net current will flow out of the device. When in equilibrium, there are no free carriers in the p and n regions, and so these regions become charged. Due to the diffusion of majority charge carriers across the interface, there is an electric field between the p-doped and n-doped region. The region of the device containing no free carriers is called the depletion region.

When a photon is incident on a pn junction, it will free an electron-hole pair from lattice if the photon has sufficient energy. If this photon is absorbed within the depletion region of the diode, then due to the electric field, the electron and hole will be separated. These electrons/holes will generate a current in the circuit connected to the diode. However, if a photon is absorbed outside the depletion region, then the electron-hole pair will not be separated and will recombine. These photons will not contribute to the current in the circuit.

To widen the depletion region, an external voltage can be used. When a *reverse bias* – a positive voltage across the cathode and anode – is applied to the photodiode, there is a larger electric field difference at the interface between the p-doped and n-doped region. This increases the volume of the photodiode in which absorbed photons will generate photocurrent.

To further increase the width of the depletion region, a layer of intrinsic semiconductor can be used to separate the p and n-doped region. An intrinsic semiconductor has no charge as it has an equal number of excited electrons and holes. An electric field will be present in a larger volume of material and so more photons will be

converted into electrons which leave the photodiode diode.

The dark noise and dark current of a photodiode has several contributors. In low bias regimes, the dark current is driven by the diffusion of minority carriers [?, 7]. However, the thermal generation-recombination of electron/hole pairs drives the dark noise at low biases [8]. Therefore, the dark current does not predict the dark noise, however as both of them increase with higher levels of defects, a diode with a high dark current will have a high dark noise. When the bias is large enough, electrons can tunnel across the the junction leading to a great increase in dark current.

InGaAs is a widely used semiconductor for making photodiodes. Typically, the indium content is around 53%, sometimes this is made clear by writing $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$, so that the InGaAs can be lattice matched to InP substrate on which the photodiode is grown on. At this level of indium doping, the InGaAs photodiode will be suffer from few mechanical defects and so the diodes created with it will be of high quality. Typically they are sensitive up to wavelengths around 1.6 μm .

Longer wavelength sensitivity can be achieved by increasing the amount of indium doping. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 2.6 μm . One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate on which its grown. Lattice mismatching can arise due to a strained crystal structure, and since the extra indium in an extended InGaAs photodiode causes this [9] it is hard to fabricate a low noise extended InGaAs photodiode [10]. Above 53% indium, a buffer layer of InGaAs with less indium in between the absorbing

layer and the substrate can be used to mitigate the lattice mismatch slightly, but the dislocations still give rise to trap states which cause the generation and recombination of charge carriers [11]. Lattice mismatching is the source of the large amounts of dark current found in extended InGaAs as well as the $1/f$ noise. Additionally, the defects that arise due to the strain caused by the lattice mismatch between the InGaAs and InP substrate effectively increase the doping level of absorbing layer of the photodiode which leads to a decrease in quantum efficiency, so in order to obtain high quantum efficiency the defect level must be low [?].

1.3 What is Needed from the Read-out Photodiode in a Gravitational Wave detector

The photodiodes used in the readout of a gravitational wave interferometer should have a quantum efficiency of more than 99%. The reason for having such a high quantum efficiency requirement is that to achieve the baseline level of ~ 10 dB of squeezed light that future gravitational wave detectors will use, losses in the interferometer have to be extremely low as losses degrade the amount of squeezing achieved.

Photodiodes for gravitational wave detectors should have low $1/f$ dark noise so that they are not limited in the detection band by electronic noise. Third generation ground based gravitational wave detectors will be quantum noise limited down to about 10 Hz [12], so the $1/f$ noise must be below the shot noise of the photo-current above this frequency.

The $1/f$ noise could be avoided by using a heterodyne detection, however the shot noise would increase if the mixing of the modulation and demodulation waveforms was not time independent [13] and this limits the choice of waveforms. These schemes recover increase the SNR by increasing the signal, however the use of squeezing means that this approach would still not be sufficient as the noise would still be enhanced compared to a homodyne readout. However, it has been proposed that the increase in noise associated with heterodyne readout could be avoided by the use of a broadband squeezer [14].

Current gravitational wave detectors operate in the 1-10 mW range at the output [15] to ensure that the junk light from other modes is a small portion of the total signal. Thus, any new photodiode should be able to produce ~ 10 mA of photocurrent.

The quantum efficiency, saturation limit, dark current and dark noise was investigated in a selection commercially available photodiodes. The photodiodes that were tested and their properties according to their datasheets¹ are listed in table 1.1. According to each of the diodes datasheets, none of them have 99% quantum efficiency; however, the quantum efficiency of a diode can be improved by increasing the bias

¹Datasheets often quote the noise equivalent power (NEP) of a photodiode. The NEP is the noise spectral density of the dark current generated by the detector divided by the responsivity of the photodiode. For the Hamamatsu g12183 series diodes, the responsivity is ~ 1 A/W and the NEP is $\sim 10^{-12}$ W/ $\sqrt{\text{Hz}}$, so the noise, as claimed by the datasheet, is $\sim 10^{-12}$ A/ $\sqrt{\text{Hz}}$. This would seem to be sufficiently low to be shot noise limited down to ~ 1 pW. However, this number does not capture the behaviour of the photodiode at illumination levels of several milliwatts where a bias is required to ensure the diode does not saturate. As the noise of these diodes is not flat (see Appendix A), NEP should be specified as a function of frequency. The noise and responsivity are both functions of reverse bias and so the NEP is a function of bias. Another quantity related the NEP is the specific detectivity; this incorporates the area and bandwidth of the photodiode with the NEP.

Photodiode	Peak quantum efficiency (%)	Cut-off wavelength ^a (μm)	Diameter (mm)
FD10D	80 ^b	2.5	1
G12183-010k	77 ^b	2.5	1
G12183-005k	77 ^b	2.5	0.5
IG26x500S4i	86 ^c	2.5	0.5
IG24x500S4i	84 ^c	2.4	0.5
G12182-005k	84 ^d	2.1	0.5
G12182-010k	84 ^d	2.1	1.0

^a λ_{EM} for $\eta = 50\%$ ^b $\lambda_{\text{EM}} = 1.9\mu\text{m}$ ^c $\lambda_{\text{EM}} = 2.0\mu\text{m}$ ^d $\lambda_{\text{EM}} = 1.6\mu\text{m}$

Table 1.1: Datasheet information for the Photodiodes that were tested. Note that none of the photodiodes has a quantum efficiency of 99%, however measurements of dark noise and dark current and investigations into their optimal operating conditions is still of interest. This selection of diodes covers a range of manufacturers, sizes and cut-off wavelengths.

voltage (see Section ??). Increasing the bias increases the dark current and dark noise of the photodiode and so for each diode a maximum reverse bias voltage was found. The benchmark dark noise used was the photodiode’s dark noise at 10 Hz; the cut-off bias was determined by the voltage where the dark noise was equal to the shot noise of a 10 mA current. Below 10 Hz other sources of noise, which rise rapidly at low frequencies, will limit the sensitivity of the Einstein Telescope [2] and Cosmic Explorer [16]. The bias voltage also affects the saturation limit of the photodiode. If a diode was found to have a low saturation threshold ($\sim 1\text{ mA}$), then the diode is not fit for use in a gravitational wave detector.

1.4 Reverse Bias and the Quantum Efficiency of an Extended InGaAs Photodiode

To demonstrate the effect that reverse bias has on the quantum efficiency of a diode, the Thorlabs FD10D was used. The FD10D was exposed to a variety of beams so that saturation effects would be visible. Two styles of measurements were used: One where the reverse bias and the response of the diode were measured at DC, and a lock-in style measurement where the quantity $\frac{\Delta\eta}{\Delta V_B}$ was measured. The quantum efficiency should be independent of frequency if the diode is operating in a non-saturating regime, and so the two styles of measurements should yield the same outcome. To prove that the increase in signal is a result of increasing quantum efficiency and not some other mechanism, for instance an avalanche effect, the noise of the photocurrent was compared to the level expected for shot noise at a bias voltage that yielded an acceptable level of dark noise (see Section ?? for details.).

1.4.1 Method

To measure the photocurrent produced by the photodiode, a transimpedance amplifier that would produce shot noise limited signals for 10 mA of photocurrent was used. Measurements of shot noise are needed to ensure that the photodiode's quantum efficiency increase is not due to an avalanche process. This circuit is shown in Appendix B.2. The photodiode circuit allowed for the reverse bias of the photodiode to be altered.

A 1550 nm wavelength laser (Pure Photonics PPCL300) was used to illuminate the photodiode (FD10D). The beam was split by a 50:50 beam splitter so that the light

could be amplitude stabilised. The in-loop diode was a high performance InGaAs photodiode (C3061GH), and the out-of-loop diode was the diode under test. The laser was amplitude stabilised by actuating on the laser diode's current. Shot noise limited measurements could be made above 1 kHz. A half waveplate and polarising beam splitter were used to control the power level of the light incident on the FD10D, and a lens was used to set the size of the beam.

The FD10D was exposed to a variety of different beams to investigate the possibility that current density related phenomena could affect the results. This was done by changing the beam size and beam power. As the FD10D is 1 mm, the maximum beam width would be around 330 μm in order for less than 1% clipping losses. The powers used were in the milliwatt range as these types of beam powers are the ones that a gravitational wave detector would operated with. The methods in this section can only be used to measure relative increases in quantum efficiency, measurements of absolute quantum efficiency require [find ref ??](#) additional apparatus.

To measure the effect that changing the reverse bias voltage has on the quantum efficiency of a photodiode, two approaches were used. By measuring the increase in photocurrent at DC it is straight forward to determined the increase in quantum efficiency. Alternatively, one can perform a lock-in style measurement. If the quantum efficiency depends on the reverse bias of the photodiode, then modulating the reverse bias, V_B at a frequency f will result in η modulating at f , and so the measurement will be of $\Delta\eta/\Delta V_B$. The input modulation $\Delta V_{\text{in}} = \Delta V_B$ and the output modulation $\Delta V_{\text{out}} \cos \theta = \Delta\eta V_{\text{TIA}}(V_B)$, where $V_{\text{TIA}}(V_B)$ is the DC signal from the photodiode at a reverse bias V_B and θ is the phase between the input and output

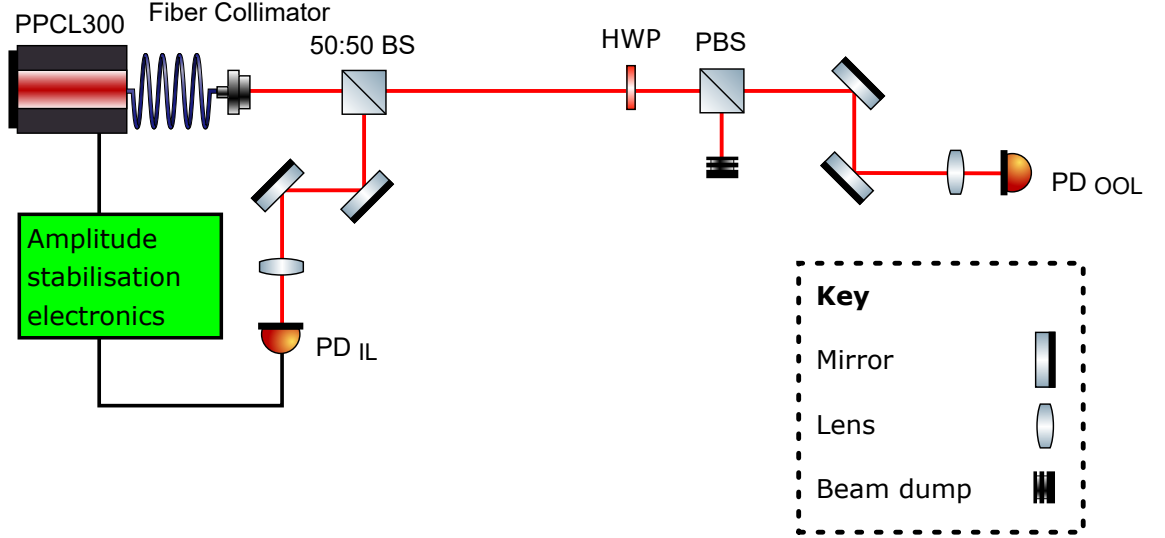


Figure 1.1: Optical layout used for measuring the effect that increasing the reverse bias of a photodiode has on its quantum efficiency and saturation limit. The out-of-loop photodiode, PD_{OOL}, was the photodiode under test. A 1550 nm laser (PPCL300) was used to test the optical properties of the photodiode. A combination of half wave plate (HWP) and polarised beam splitter (PBS) allowed for the power of the beam to be set and a lens was used to set the size of the beam on the photodiode under test. Amplitude stabilised light was needed to ensure the increase in photocurrent seen when the bias increased was due to an increase in quantum efficiency. The photodiode circuit (see Appendix B.2) allowed for shot noise limited measurements of currents around 10 mA and for the bias to be selected.

modulation.

As 0.6 V reverse bias was the highest bias that gave an acceptable level of dark noise (see Figure 1.5), the noise of the photocurrent was measured at a region where the noise of the laser was limited by shot noise. This was done to ensure that the increase in photocurrent was due to a real increase in quantum efficiency and not due to another mechanism e.g. an avalanche process. The expected level of noise, n , of the photocurrent, in units of $A_{\text{rms}}/\sqrt{\text{Hz}}$, is

$$n = \sqrt{2qI_{\text{OOL}} + 2qI_{\text{IL}} + n_{\text{IL dark}}^2 + n_{\text{OOL dark}}^2 + n_{\text{loop}}^2 + n_{\text{analyser}}^2} \quad (1.1)$$

where q is the charge of the electron, $I_{\text{OOL/IL}}$ is the current through the out-of-loop and in-loop photodiodes, $n_{\text{OOL/IL dark}}$ is the dark noise of the out-of-loop diodes and in-loop photodiode, n_{loop} is the electronic noise of the loop and n_{analyser}^2 is the noise of the analyser. By tracking the height of an intensity modulation in the light for a series of powers up to 16 mW, the linearity of the diode could be investigated.

At this bias, the quantum efficiency was calibrated using two other power meters. The power incident onto the photodiode was measured and the power reflected by the window at small angles was measured.

1.4.2 Results

The effect of increasing the reverse bias on the quantum efficiency of the FD10D is shown in Figure 1.2. For all beams intensities that the photodiode was exposed to, the increase in quantum efficiency was $\sim 15\%$ as the reverse bias went from 0 V

to 1.8 V, the stated maximum reverse bias in the datasheet of the FD10D. There may be a saturation related effect at the 1% level, beams with a lower intensity experienced a greater increase in η than beams with a higher intensity, however the quantum efficiency may not be uniform over the diode – near the edges η is greater. The difference between the DC and AC response could only be described by a non-linear process. These effects are small, and Figure 1.2 gives a clear incentive to increase the bias voltage.

Figure 1.3 shows the photocurrent measured has the noise level predicted by Equation 1.1, meaning that the increase in photocurrent at a reverse bias voltage of 0.6V is, without doubt, an increase in quantum efficiency. This measurement was performed with 4 different photocurrents, ranging from 7.5 mA to 15.5 mA, and over this range there was a 2% change in the ratio between the DC and AC measurement, this is shown in Figure 1.4. The SNR of this measurement is limited to the 1% level due to the difference between the calibration peak applied at 1111 Hz and the level of shot noise.

The internal quantum efficiency was measured using two power meters to be 70%. There is a significant amount of light reflected by the FD10D's window; if the window is not removed, then at 0.6 V reverse bias the quantum efficiency would be 60%.

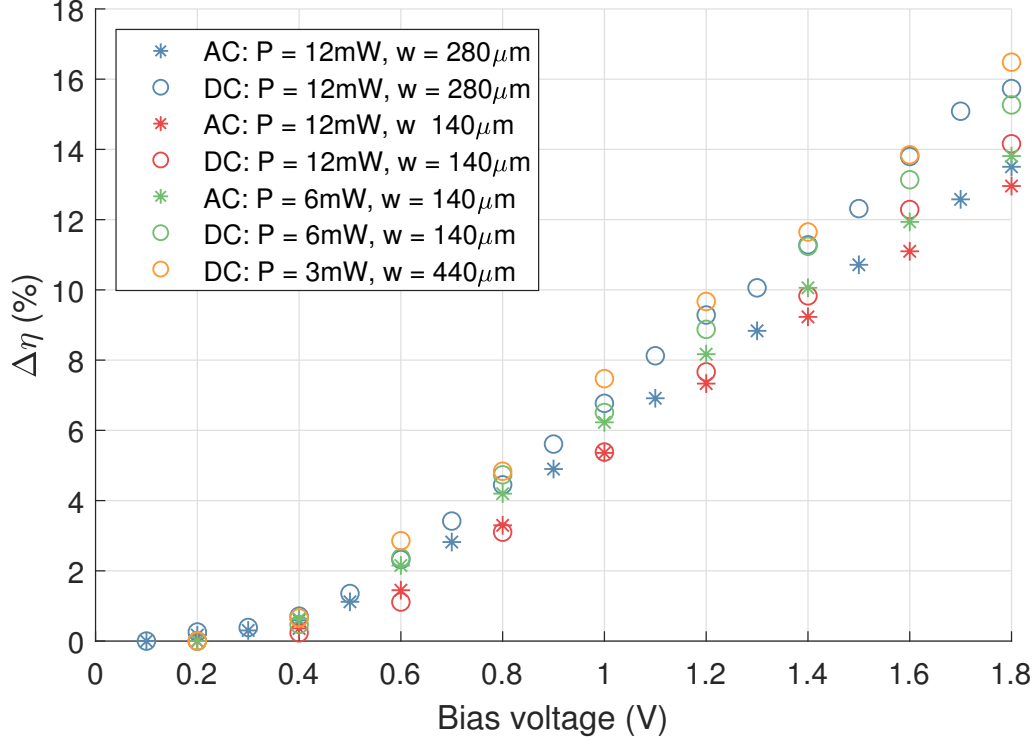


Figure 1.2: The FD10D’s quantum efficiency depends on its reverse bias; the increase in quantum efficiency was at the $\sim 15\%$ level when the bias was increased to its maximum datasheet value. Two types of measurement were used to determine the increase in quantum efficiency as the bias was increased. The first was to measure the DC response and the corresponding results are shown as circles. The second method involved modulating the bias and measuring the resulting modulation in the photocurrent, i.e. an AC measurement, and these are shown by the stars. Each set of measurements was done with a different intensity by changing either the beam’s width or power.

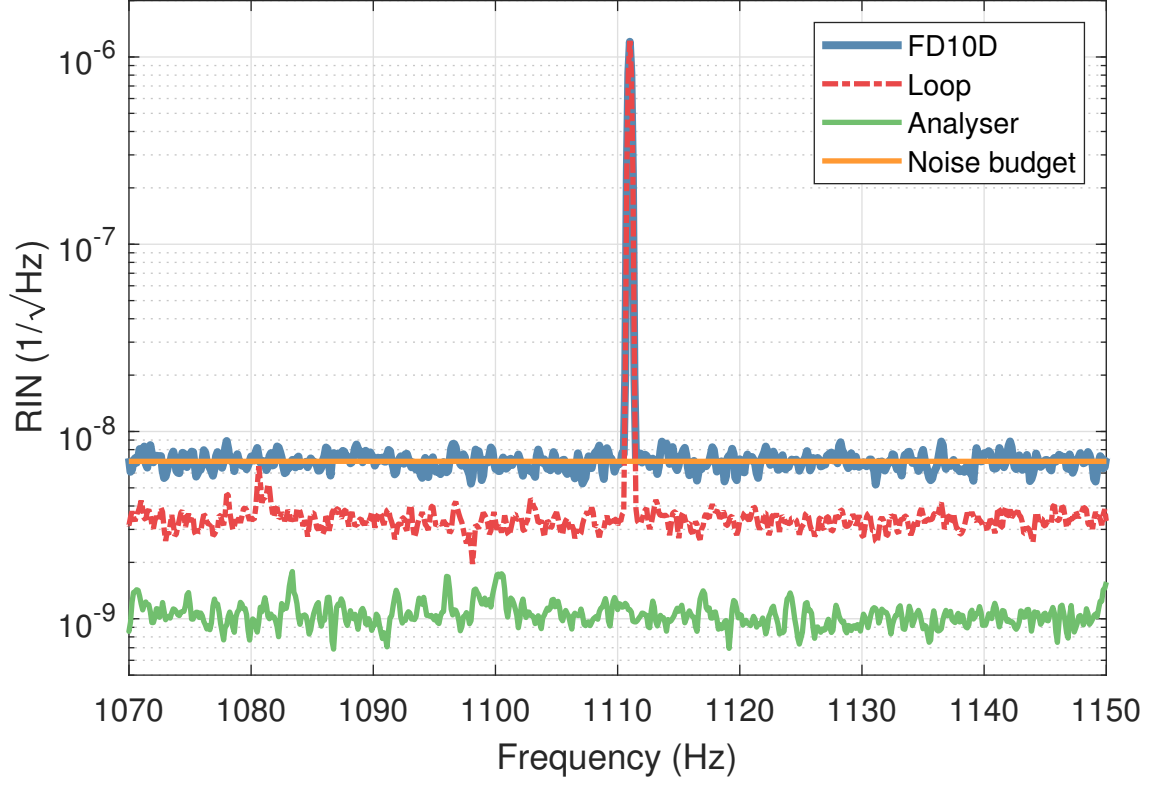


Figure 1.3: The bias of the FD10D was set to 0.6 V and it was exposed to light **find values**. The noise of the photocurrent produced by the FD10D (blue line) corresponds to the Equation 1.1 (orange line) meaning that that the quantum efficiency of the FD10D did increase rather than the increase in photocurrent being the result of an avalanche process. The in-loop signal (red line) was used to cross-check the calibration of the measurement.

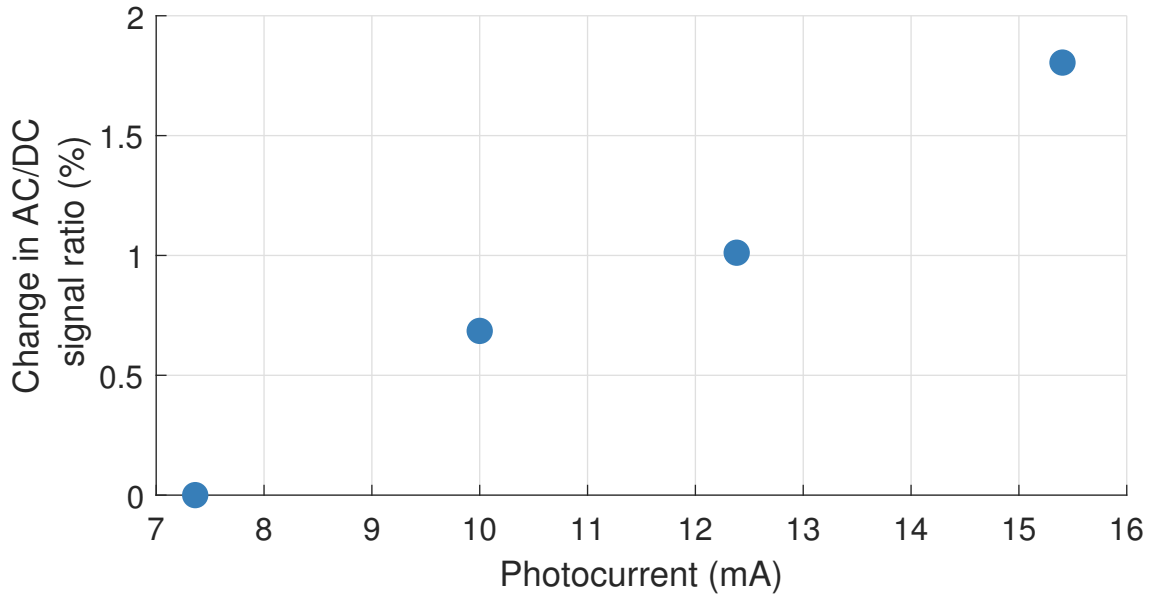


Figure 1.4: The bias of the FD10D was set to 0.6 V and it was exposed to a series of different power levels to check at what photocurrent the diode would saturate at. This was done by measuring the height of the calibration peak; if this peak decreased in size relative to the DC signal then this would indicate the AC signal was saturating. This peak could only be measured to within 1%, and so the FD10D showed no sign of saturation under these conditions.

1.5 Dark Current and Dark Noise in Extended InGaAs Photodiodes

The dark current and dark noise of extended InGaAs photodiodes has been a limiting factor in previous quantum optics experiments [4, 5]. The dark current of these photodiodes depends on their reverse bias. The larger the reverse bias, the higher the dark current. The dark current of extended InGaAs photodiodes is high because of the relatively large amounts of dislocations due to lattice mismatch between the InGaAs and the substrate that it is grown on.

There is a balance to strike between the quantum efficiency and dark noise. It is desirable to have as large a reverse bias as possible as this increases the quantum efficiency and saturation limit of the photodiode (see section ?? or figure ??). The dark noise of the photodiode should be below the shot noise level of the light that will be detected. As a bench mark, the dark noise of the photodiodes at 10 Hz is compared to the dark noise of 10 mA of photocurrent.

The noise of the current made by the photodiode can be modelled as

$$i_{\text{rms}} = \sqrt{2qi + \frac{4k_B T}{R_{\text{PD}}} + \left(\frac{i_t}{f^\beta}\right)^2}. \quad (1.2)$$

The first term under the square-root, $2qi$, is the shot noise associated with the DC current i . The second term, $\frac{4k_B T}{R_{\text{PD}}}$ is the Johnson-Nyquist noise due to the shunt resistance of the photodiode, R_{PD} ; this term can become significant when the shunt resistance is low as it is in some extended InGaAs photodiodes. The final term is is Hooge's law of noise in semiconductors [8]; i_t is the characteristic noise level and β

is usually 0.5 ± 0.1 .

To measure the dark noise produced by the extended InGaAs photodiodes, a transimpedance amplifier with high gain was used. The photodiodes were placed in a light tight box while the measurement was being made.

1.5.1 Room Temperature

The dark noise as a function of bias was measured at room temperature ($\sim 20^\circ\text{C}$) for the photodiodes listed in Table 1.1 . The results for the FD10D are shown in Figure 1.5. The dark noise for the rest of the diodes can be found in Appendix A.1. Not all diodes had a simple noise spectrum; however, usually, the spectrum would fall as $1/f^{0.5}$.

The dark noise at 10 Hz and the dark current as a function of bias for each diode is shown in Figure 1.6. The noise at 10 Hz was either determined by reading the value from the corresponding graph in Appendix A.1 or, if more precision was required, by fitting a $1/f^\beta$ slope to the the data and inferring the noise from the fitted line.

Where there were duplicate diodes to choose from, the superior diode was selected for characterisation. There can be a significant amount of variation in quality within a batch of photodiodes. Figure 1.7 shows there was two orders of magnitude difference in noise spectrum of two ‘identical’ Hamamatsu photodiodes.

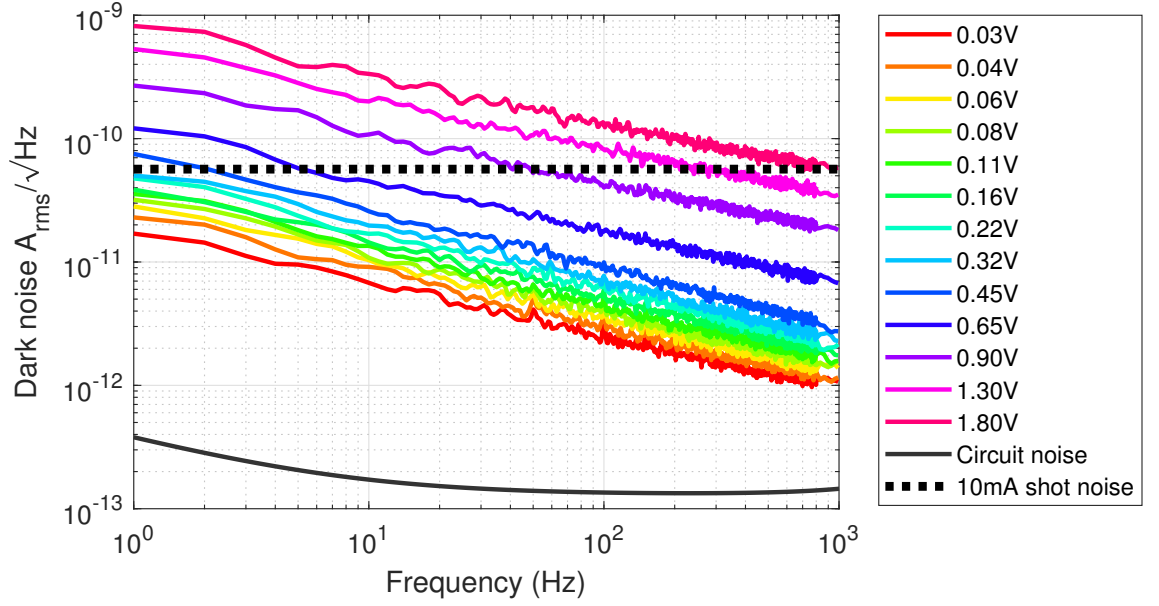


Figure 1.5: The dark noise of the FD10D increases as the reverse bias is increased. Above 0.6 V, the dark noise of the FD10D would limit the noise of a 10 mA current. As the noise spectral densities have no knees, the function i_t/f^β can be fitted to the it. See appendix A.1 for the dark noise spectra of other photodiodes. Below 0.9 V reverse bias, the spectra fall as $f^{-0.45}$ which is typical for ‘1/f’ noise. Above 0.9 V bias the spectra fall as $f^{-0.4}$. This data is shown along side the results for the other diodes tested in Figure 1.6.

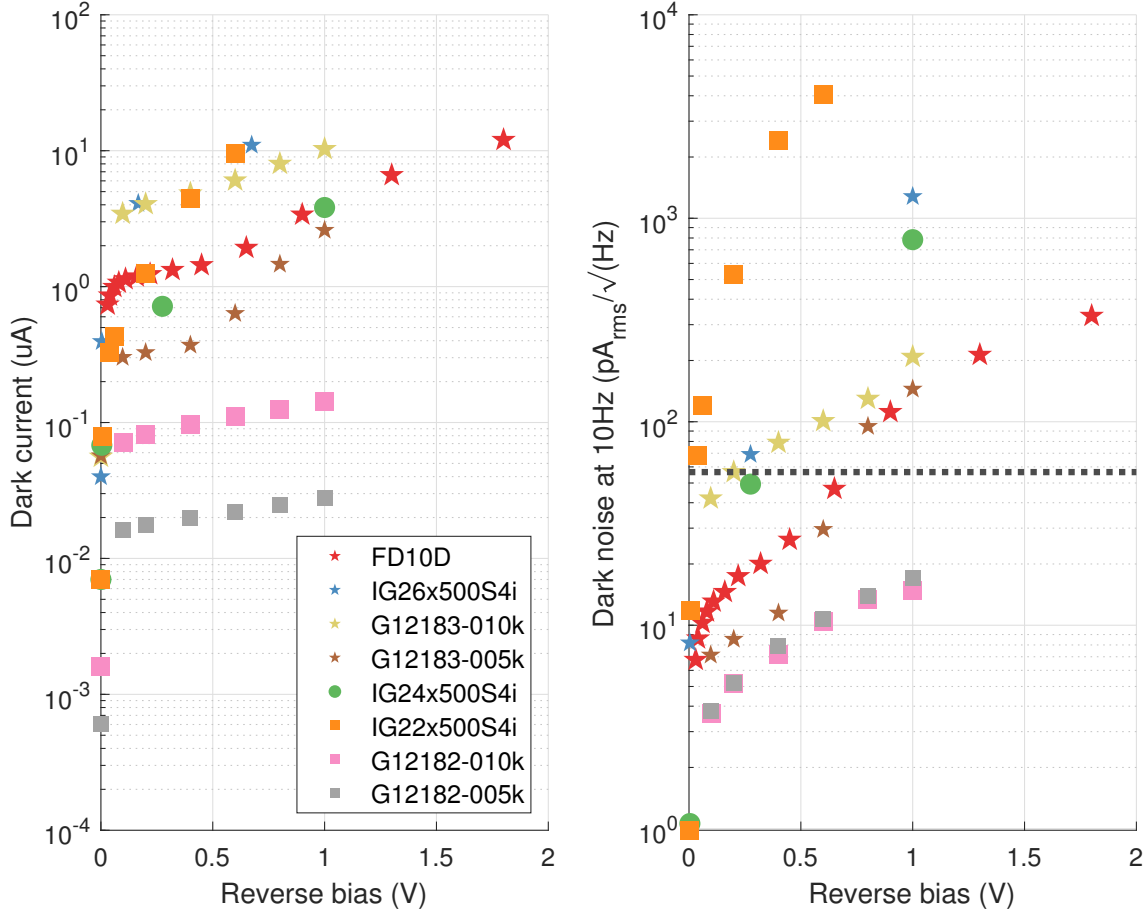


Figure 1.6: Left panel: the dark current for each diode was measured. Diodes of a similar cut-off wavelength share the same marker shape. The 1 mm photodiodes have larger markers than the 0.5 mm photodiodes. Right panel: the dark noise at 10 Hz for each diode. The dashed line indicates the level of shot noise for a 10 mA current.

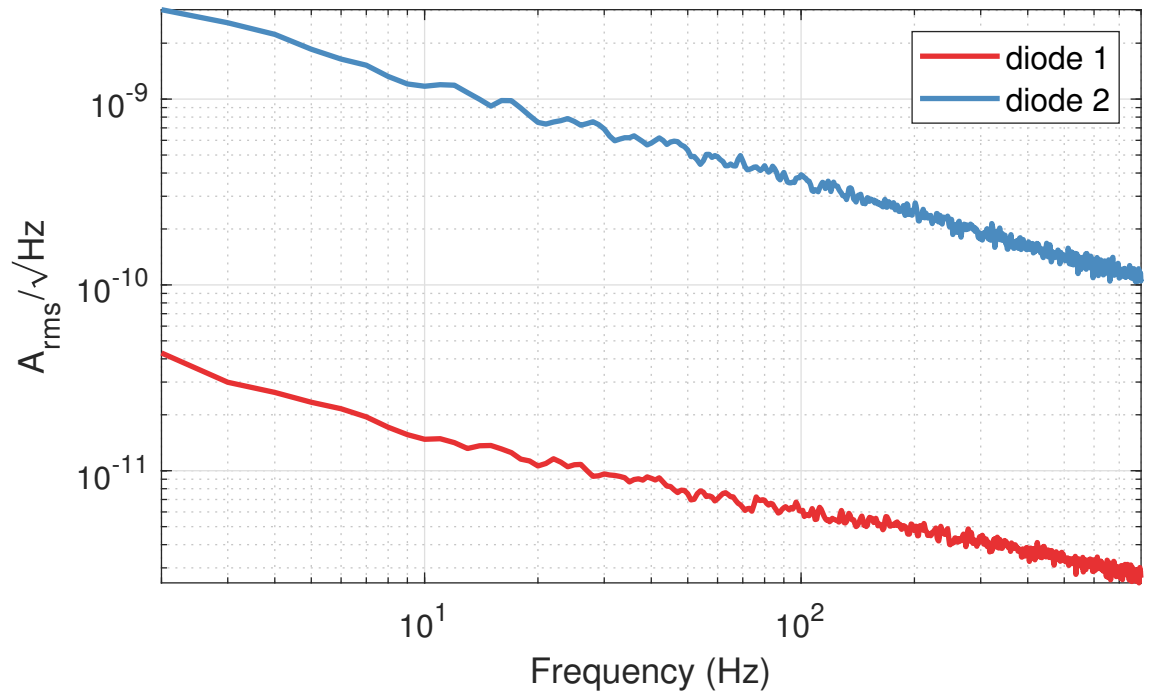


Figure 1.7: The difference in dark noise between two Hamamatsu photodiodes (g12182-010k) of the same kind under identical operating conditions (room temperature at 1 V bias). Diode 2 performs nearly 100 times better than diode 1.

1.5.2 Cooling Experiment

The typical low temperature damage threshold for extended InGaAs photodiodes is -40°C so the dark noise of the photodiodes listed in Table 1.1 was investigated over a range of temperatures from -40°C to 30°C . To provide controlled cooling to the photodiode, the photodiode was glued (using Cryo-Varnish) to a block of aluminium, shown in figure 1.8. This block had a copper rod with a diameter of a few mm soldered to it, and this rod was inserted into a liquid nitrogen dewar. The block of aluminium had power resistors to provide heating so that temperature of the diode could be controlled. A PID loop was implemented using a pt1000 temperature sensor and a multimeter (keithley 2000) configured to make a 4 wire resistance measurement. A Labview program was used to perform the PID calculation. This program read in the temperature signal and calculated the required voltage to be generated by a programmable power supply (Temna 72-2540) connected to the power resistors that heated the aluminium block.

The dark noise and dark current for each diode in Table 1.1 was measured at the diode's maximum reverse bias specified in the data sheet. This is shown in figure 1.9 and the individual spectra are shown in Appendix A.2. Except for the IG24x500S4i, the dark noise and dark current for each photodiode fell exponentially as the photodiode was cooled.

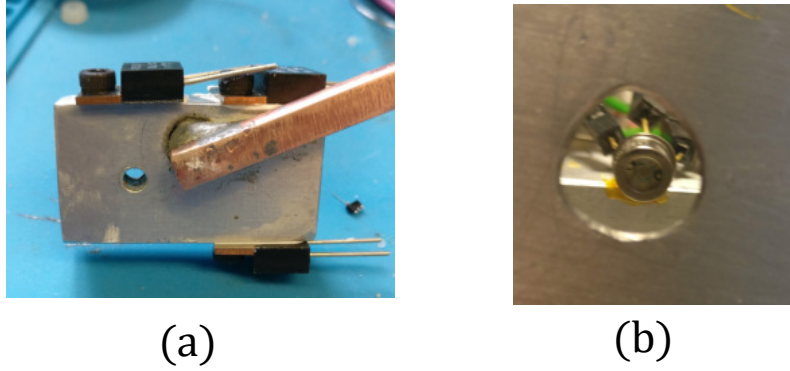


Figure 1.8: (a) The photodiode was affixed to this aluminium block. The power resistors provided heating to the block and the copper rod provided a thermal link between the liquid nitrogen dewar and the block. (b) A photodiode glued to the block such that a large proportion of the photodiode was in contact with it.

1.6 The Saturation Limit of Some Laser Components Extended InGaAs Photodiodes

The laser components diodes were found to saturate at low powers if they were biased with a low enough voltages such that their dark noise was below the noise target. The laser components photodiodes were exposed to a light signal of $RIN = 1.2e-6^2$. A similar type of measurement is reported in [17]. This could be cross-checked by measuring the light modulation on the high performance in-loop photodiode. The power of the light was increased, and the decrease in the relative intensity of the AC modulation was recorded. This is shown in Figure 1.10. A decrease in AC signal indicates that the diode is saturating.

²The peak cannot be expressed in spectral density units, however the height of the out-of-loop peak just needs to be compared to the in-loop sensor's measurement of the peak.

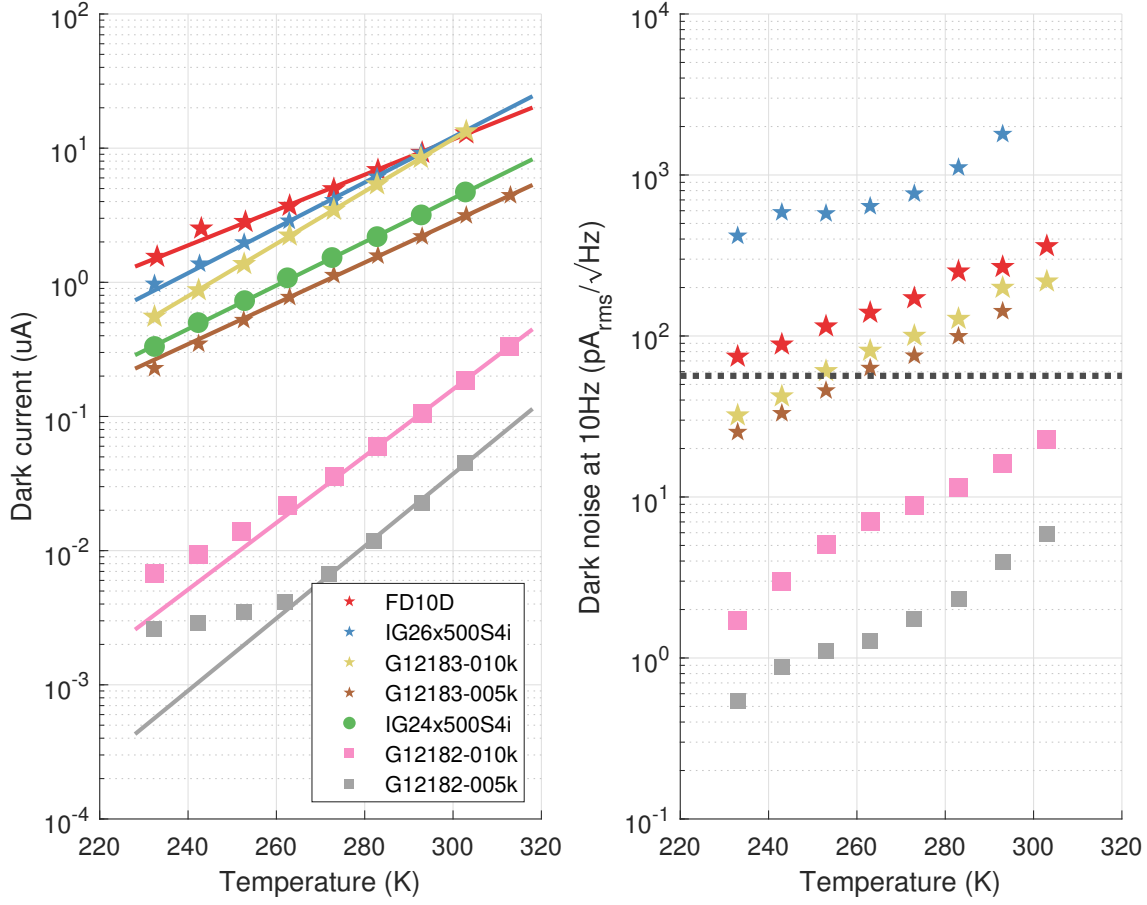


Figure 1.9: Left panel: the dark current of each photodiode at their maximum datasheet specified reverse bias as a function of temperature. The apparatus limited dark current measurements to a couple of nanoamps. Exponentials were fitted to the data. Diodes with similar cut-off wavelengths share marker shapes, and 1 mm diodes have larger markers than the 0.5 mm ones. Right panel: the dark noise of each diode was measured as a function of temperature. The black dashed line indicates the shot noise level for a 10 mA current. At 10 Hz, the measurements for the IG24x500S4i did not change much as the temperature changed, there was a knee around this frequency (see Figure A.13); the noise of the IG24x500S4i was $600 - 700 \text{ pA}/\sqrt{\text{Hz}}$ at 10 Hz.

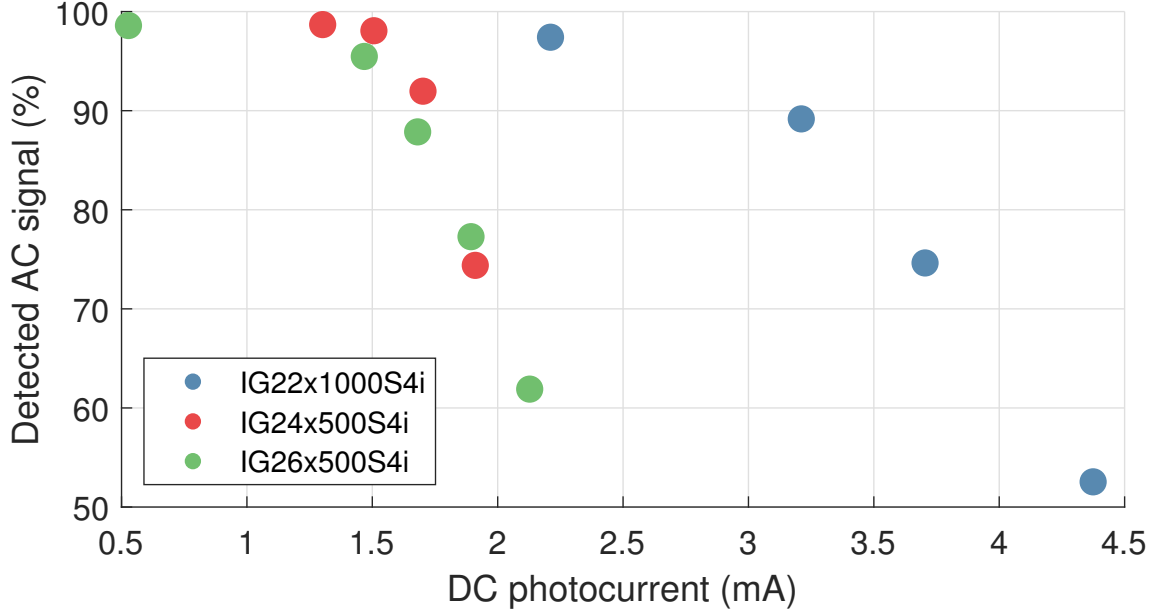


Figure 1.10: The saturation limit of three Laser Components photodiodes was measured. The photodiodes were biased so that their noise at 10 Hz was equal to the shot noise of a 10 mA current. The IG24x500S4i and IG26x500S4i were exposed to a beam whose 3σ width was 220 μm and the IG22x1000S4i was exposed to a beam whose 3σ width was 1000 μm . The size of the a light modulation whose relative size was known was tracked as the DC photocurrent was increased; the more saturated a photodiode is, the smaller the amount of signal was detected. These diodes had low saturations limits compared to the FD10D, which showed no signs of saturation up to 16.5 mA (see Figure ??).

1.7 Excess Noise in the IG24x500S4i

The experiment explained in Section ?? showed that the noise of the photocurrent IG24x500S4i was higher than predicted by equation 1.1 by a factor of ~ 2 . The diode was not saturating during this measurement as the 1111 Hz peak has the right relative intensity.

To rule out bias noise causing this, a noise projection measurement was made. The transfer function between the bias at the anode of the photoiode and the output of the transimpedance amplifier was measured while 1 mA of photocurrent was being generated by the diode. A calibration modulation was applied to the bias voltage, and the noise at the anode of the diode and the noise at the output of the transimpedance amplifier were simultaneously measured. Using this method, 42.5 dB of bias voltage noise was removed from the measurement; however, this did not remove the excess broadband noise. This is shown in figure 1.12.

The ideality factor of the IG24 was investigated. The equation of a diode is

$$I = I_p + I_s \left(\exp \left(-\frac{qV_b}{nk_B T} \right) - 1 \right), \quad (1.3)$$

where I_p is the photocurrent, I_s is the saturation current of the diode, q is the charge of the electron, V_b is the bias of the diode, n is the ideality factor, k_B is Boltzmann's constant and T is the temperature of the diode. If n differs significantly from 1, then it indicates that additional behaviour that was not described in Section 1.2 occurring within the diode. The current flowing through the diode as a function of forward bias was measured and is shown in figure 1.13. The SciPy python function

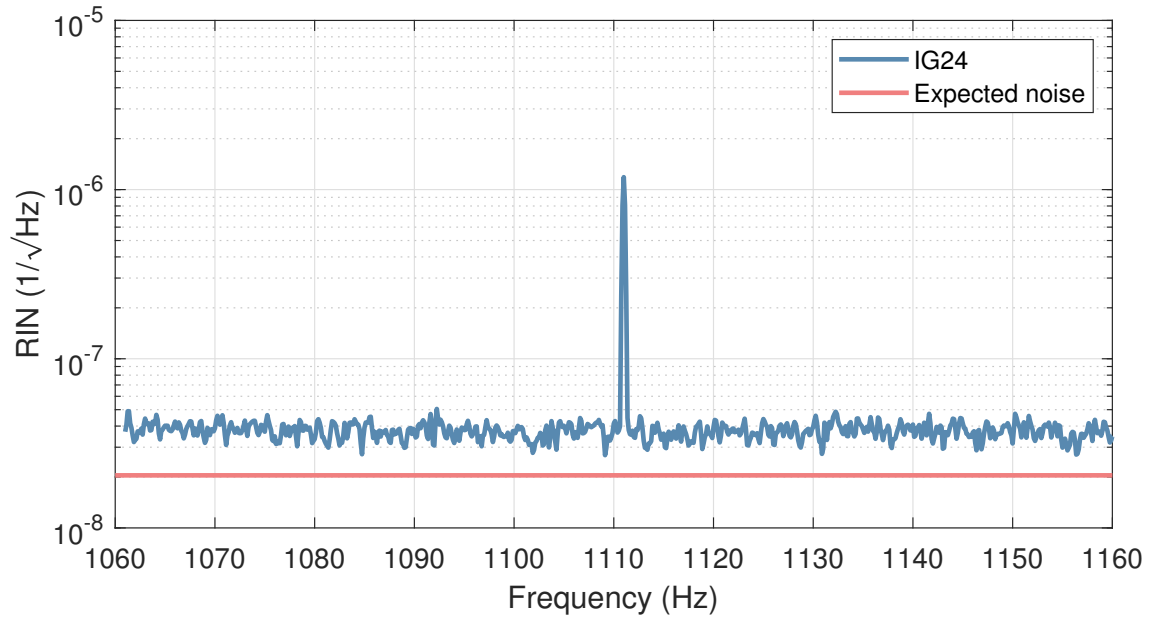


Figure 1.11: The noise of a 1.3 mA of photocurrent made by the IG24x500S4i (blue) exceeded the expected amount (red) based on Equation 1.1 by a factor of 1.75. This beam was not saturating the diode as the height of the 1111 Hz signal was the the right level. This excess noise was also found at 0.25 mA of photocurrent.

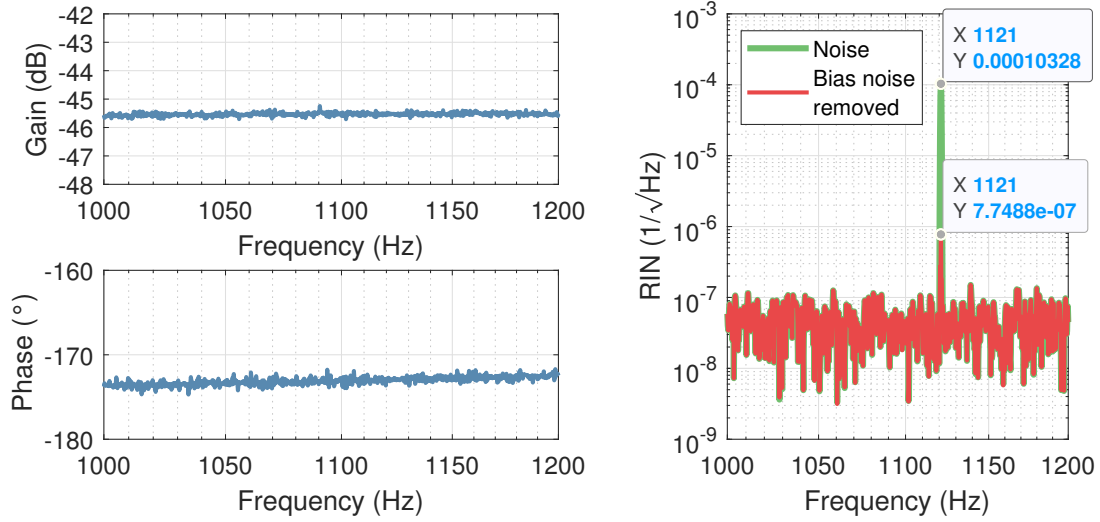


Figure 1.12: A noise projection measurement was performed to determine if the excess noise in the photocurrent produced by the IG24x500S4i (see Figure 1.11) was due to bias noise. The left hand panels show the bode plot of the transfer function between the photodiode's cathode and the output of the transimpedance amplifier when the photodiode was producing 1.3 mA of photocurrent. The right hand panel shows the effect of subtracting bias noise from the signal at the output of the transimpedance amplifier. The peak at 1121 Hz was a modulation applied to the photodiode's cathode; 42.5 dB of this signal was removed by using the noise projection technique which is in close agreement with the amount expected from the measured transfer function. However, the level of the broadband noise remained unchanged thus indicating that the excess noise witnessed in the photocurrent produced by the IG24x500S4i was not due to bias noise.

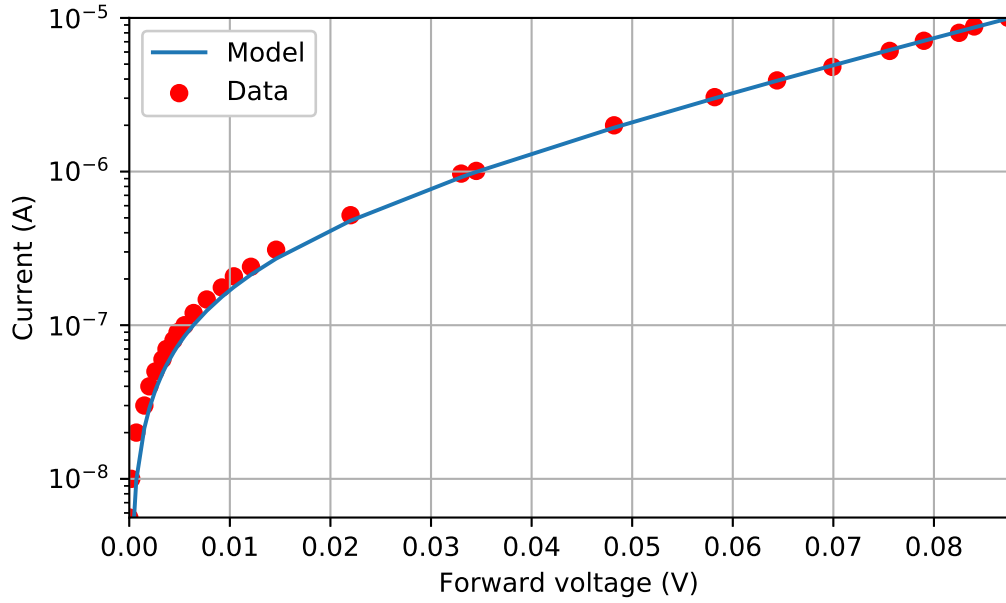


Figure 1.13: To investigate possible reasons why there was excess noise in the photocurrent produced by the IG24x500S4i, its ideality factor was measured (red points). Equation 1.3 was used to do an unweighed fit to the data ($I_0 = (360 \pm 10)$ nA and $n = 1.02 \pm 0.01$). The IG24x500S4i was found to be close to ideal, as n was close to 1, and so this does not provide clues to the source of excess noise in the photocurrent.

`scipy.optimize.curve_fit` was used to do an unweighted fit of the data to Equation 1.3. The diode was found to have $n \simeq 1$. To verify Equation 1.3, light was shone onto the diode. This is shown in figure 1.14.

The source of this excess noise is still unclear: the diode was not saturating as the calibration signal was the right level relative to the DC photocurrent, a noise projection measurement showed that the bias voltage noise was not the source of

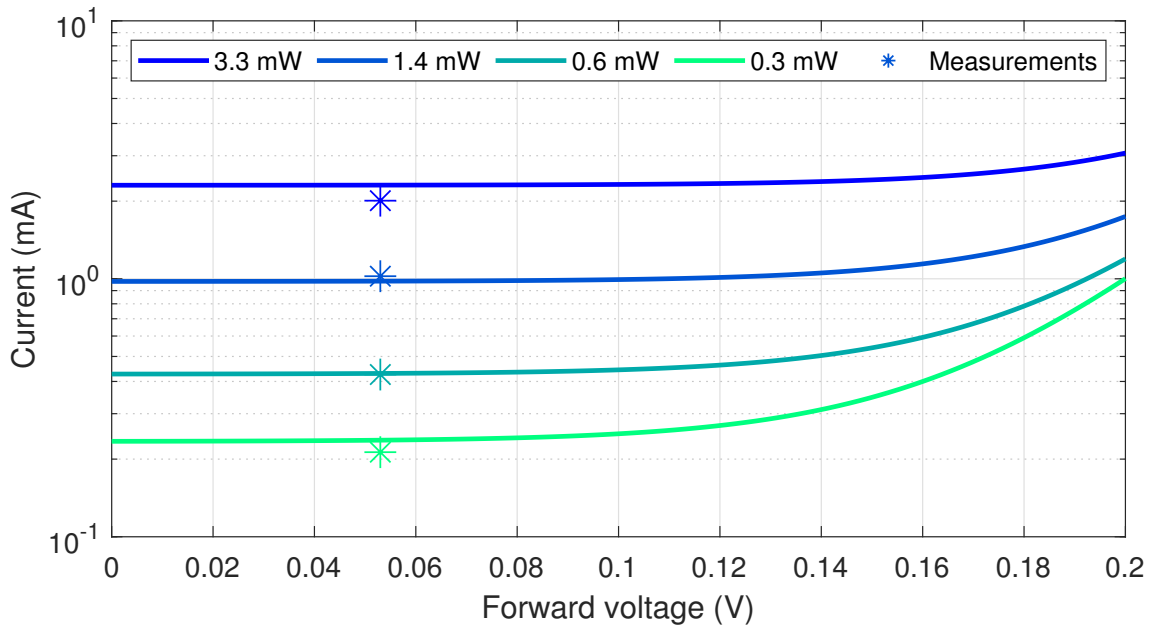


Figure 1.14: Different levels light levels were shone onto the photodiode and the current produced at 0.05 V reverse bias was recorded. These measurements are shown by the stars. Based on the model obtained from the data shown in Figure 1.13, the current flowing through the diode was predicted (indicated by the lines). The model and the measurements are in agreement with each other although the diode may have been saturating for the 3.3 mW measurement.

the excess photocurrent noise, and a measurement of the ideality factor was used to determine if that the photodiode behaved in as an ideal photodiode would. The level of noise cannot be explained by the Johnson-Nyquist noise of the shunt resistance of the photodiode as the excess noise is far too high.

1.8 Properties of an InSb detector

InSb detectors have high quantum efficiency [6] to short infrared wavelength light. A P5968 detector was tested, and the noise of the current produced by it was investigated. InSb detectors must be operated at low temperatures so liquid nitrogen was used to cool the photodiode to 77 K. The transimpedance amplifier used to measure the photocurrent is shown in Appendix B.1.

The detector was characterised at by placing a a low emissivity material in front of the detector. The InSb is sensitive to wavelengths of up to be 5.3 μm . Based on the solid angle covered by the material as seen by the detector, the expected level of thermal radiation that would be detected could be calculated. This expected photocurrent was 5.5 μA . The DC current from the device was measured to be 5.3 μA . The results from this test are shown in Figure 1.15. This corresponds to the device having a quantum efficiency of $\sim 90\%$.

As the quantum efficiency of InSb photodetectors depends strongly on the reverse bias for 1.55 μm wavelengths [18], the behaviour of the P5968 under a reverse bias was investigated. A reverse bias was applied to the device and this generated $1/f$ noise. The full data is shown in Section A.3. The noise of the photocurrent also showed a broadband increase, this is shown in Figure 1.16.

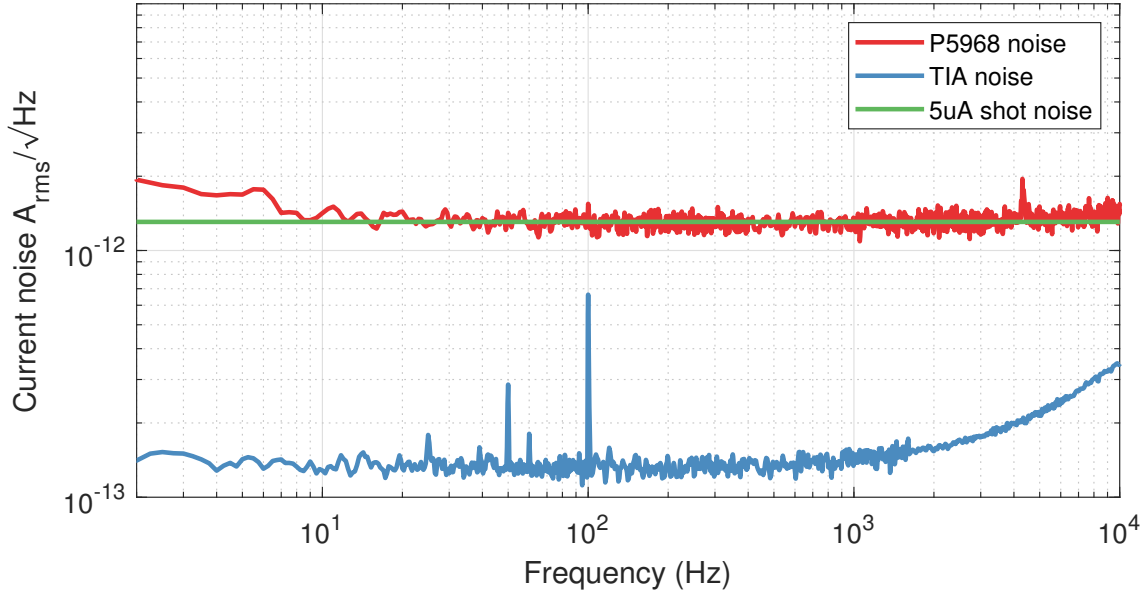


Figure 1.15: The P5968 had its cathode grounded and the noise of the photocurrent due to the blackbody radiation produced by room temperature aluminium was measured (red). The noise level of the photocurrent corresponds to shot noise (green). The noise of the transimpedance amplifier (blue) is well below the noise of photocurrents noise.

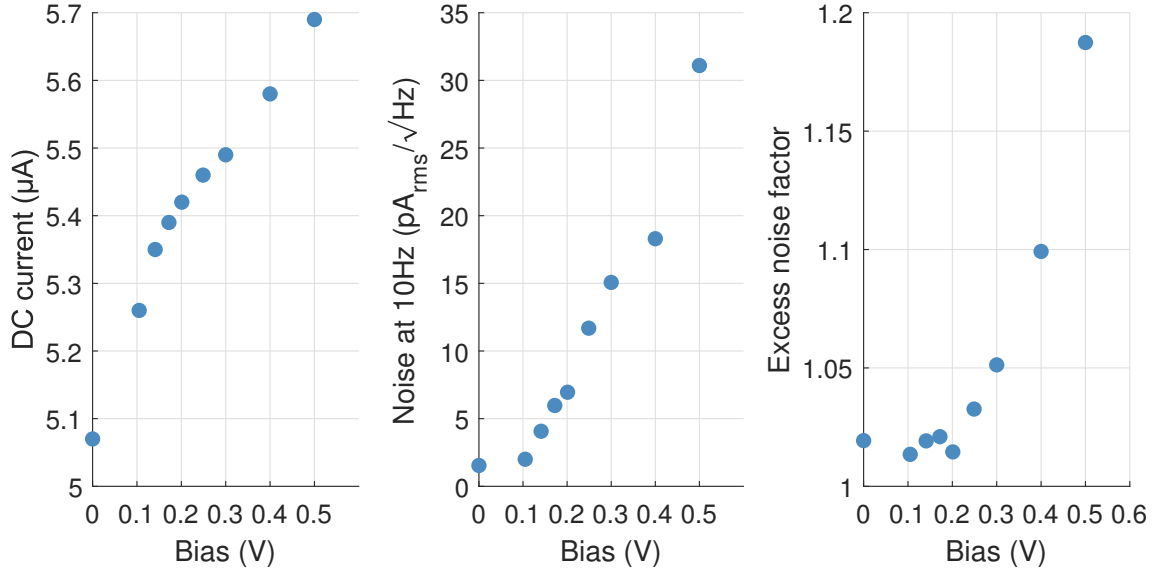


Figure 1.16: Left panel: the DC dark current of the P5968 as a function of bias. Middle panel: $1/f$ noise rose as the bias of the P5968 was increased (see Figure A.15). Right panel: At a frequency where $1/f$ was not present, the noise of the photocurrent was measured. The excess noise factor is the multiplicative factor that the shot noise corresponding to the DC photocurrent needs to be multiplied by get the noise level that was measured.

1.9 Discussion

None of the photodiodes tested in this Chapter would meet the requirements described in Section 1.3 as the quantum efficiency is specified in all of their datasheets to be below 99%. Communication with several commercial manufacturers of extended InGaAs photodiodes suggest that there is no off-the-shelf photodiode capable of meeting the requirements set out in Section ??; these results support and give insights into the best possible operating conditions for an off-the-shelf extended InGaAs photodiodes.

The results cover seven different models of extended InGaAs photodiode and so it is possible that another photodiode manufacturer (e.g Teledyne Judson) have better performing photodiodes than the ones tested in this Chapter. Additionally, as a relatively small number of photodiodes have been tested, with only two copies of most photodiodes available for testing, batch-level statistics cannot be obtained from the results. For instance, FD10D reported on in this chapter has a lower noise than the G12183-010k reported on in this chapter; however, this does not tell one if FD10Ds are generally better than G12183-010ks.

Section 1.4 shows the effect that biasing an extended InGaAs photodiode can have on its quantum efficiency. The quantum efficiency of the FD10D was increased by 15% when its bias was set to its maximum value. This increase in quantum efficiency occurs because the electron-hole pair created spend less within the depletion region, thus they have a smaller chance of recombining with another hole/electron. This motivates using increasing the bias of a photodiode as the quantum efficiency can be greatly increased.

The depletion region of Extended InGaAs photodiodes is quite small when there is no bias because it is difficult to create thick Extended InGaAs wafers due to the lattice mismatch between the InP substrate and the InGaAs. Lattice mismatches strain the InGaAs and generates dark current and dark noise as well as reduces the quantum efficiency of the diode; the diode can be made thinner to reduce this strain, although this reduces the quantum efficiency. High quantum efficiency is only required in experiments where non-classical states of light are being detected; usually, if a bigger signal is required the gain can be increased by using a larger resistor in the feedback path of the transimpedance amplifier or by using an avalanche photodiode.

The high amount of dark noise associated with the bias that was found in the Laser Components range of diodes (see section ??) meant that they had to be operated with low biases if they were to be able to produce shot noise limited photocurrents above 10 Hz. This resulted in them having a low saturation level, see Figure 1.10. This is another reason why biasing a photodiode may be necessary. To avoid this saturation limit, one could split the beam and use an array of diodes to detect the light. However, better performing photodiodes could be used instead; the FD10D showed no signs of saturation, see Figure 1.4, when operating with a bias that fulfilled the noise criteria.

Usually, applying a reverse bias is done to increase the bandwidth of a photodiode, however a few hundred picofarads of capacitance is manageable amount to design a transimpedance amplifier around for a DC readout scheme such as that used by LIGO. For instance, the photodiode circuit used in section 1.4 (see Appendix B.2) was quiet enough to be limited by the shot noise of a 10 mA photocurrent.

While the dark current does not depend strongly on the reverse bias when the bias is below the tunnelling threshold, the dark noise of every photodiode increased as its reverse bias increased. Although photodiodes with higher dark current tended to have higher dark noise, the dark noise could not be predicted from the dark current. For instance, the FD10D had a larger dark current but lower noise than the IG24x500S4i. The dark current tended to be greater in larger photodiodes, this is due to larger diodes being more likely to contain defects. However, the Laser Components photodiodes break this rule as they were all of low quality. The Hamamatsu photodiodes with shorter cut-off wavelengths (less indium), had lower dark current and dark noise compared to the photodiode with longer cut-off wavelengths.

The dark noise can vary considerably amongst a batch of photodiodes, for example there was nearly a factor of 100 between the dark noise of the two Hamamatsu G12182-010k tested (see Figure 1.7). Whether this is due to their manufacturing or how they were handled prior to purchase is unclear, however this result shows that it may be worth buying a batch of Extended InGaAs diodes and selecting the best ones from it.

The dark noise of a photodiode can be significantly reduced by cooling it. The dark current and dark noise of a photodiode increases exponentially with temperature. Photodiodes can be purchased with thermoelectric coolers but integration of such a photodiode into a vacuum system was not investigated and could pose an engineering challenge.

1.10 Conclusion

Extended InGaAs photodiodes are widely available detectors that can be used to detect $2\text{ }\mu\text{m}$ light; however, off-the-shelf detectors do not fulfil the requirements of a photodiode that can be used in the readout of a gravitational wave detector. This is due to gravitational wave detectors needing high quantum efficiency photodiodes so that the shot noise of the signal is not significantly enhanced.

It is technically challenging to create Extended InGaAs photodiodes with high quantum efficiency due to the lattice mismatch between the InP substrate and the InGaAs. While the depletion region of a photodiode can be made larger by running the photodiode with a reverse bias, the dark noise of an Extended InGaAs photodiode increases when reverse biased so dark noise measurements of a selection of photodiodes were made to investigate suitable operating conditions for them.

To overcome the manufacturing challenges associated with creating extended InGaAs photodiodes, a dedicated wafer would need to be commissioned. Increases in quantum efficiency can be obtained by fabricating photodiodes to have antireflective coatings however challenges associated with creating high quantum efficiency photodiodes lies in making is thick enough to absorb 99% of the incident photons. Reference [19] shows results from a photodiode with 95% quantum efficiency and a cut-off wavelength of $2.4\text{ }\mu\text{m}$, however due to the exponential dependence of absorption on thickness, it is still a significant challenge to reach 99% quantum efficiency.

InSb detectors may also be useful as they have near unity quantum efficiency at short infra-red wavelengths, however their quantum efficiency below $2\text{ }\mu\text{m}$ can exceed

unity due to multi-electron processes [6]. An InSb device was tested however due its characteristics at $2\text{ }\mu\text{m}$ are still unclear, to investigate these the device's window would need removing and a $2\text{ }\mu\text{m}$ laser would needed.

Alternative photodiode materials could be used to detect $2\text{ }\mu\text{m}$ light. There have suggestions that HgCdTe could be used to make a photodiode with 99% quantum efficiency at $2\text{ }\mu\text{m}$ as HgCdTe does not suffer from strain related problems, however the linearity and dark noise of such HgCdTe photodiodes is unknown.

Appendix A

Extended InGaAs Diodes Noise Spectrums

A.1 Bias Section

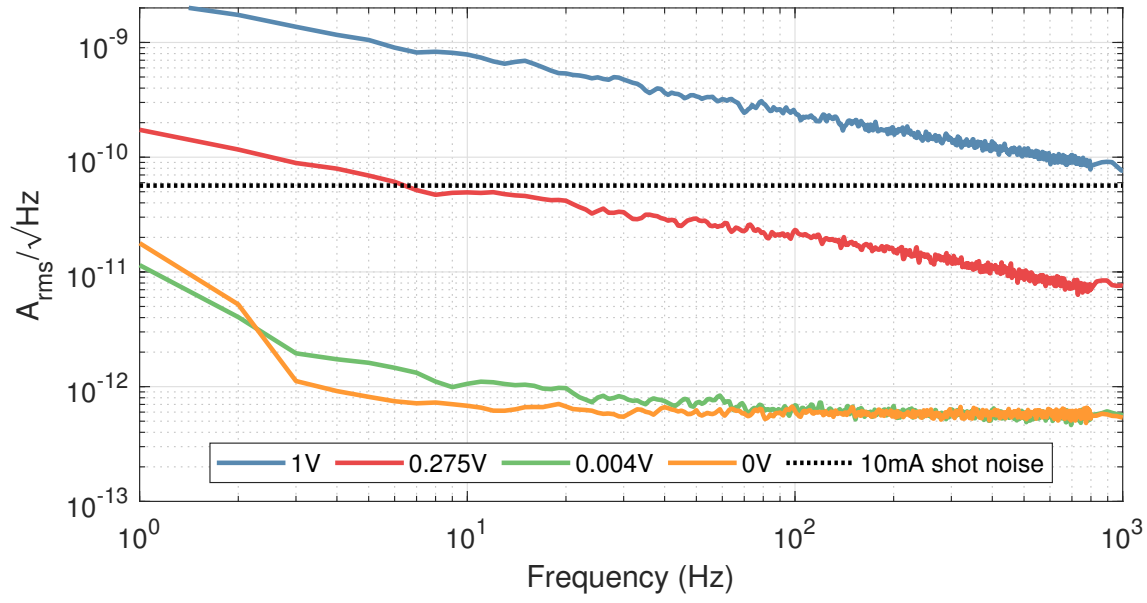


Figure A.1: IG24

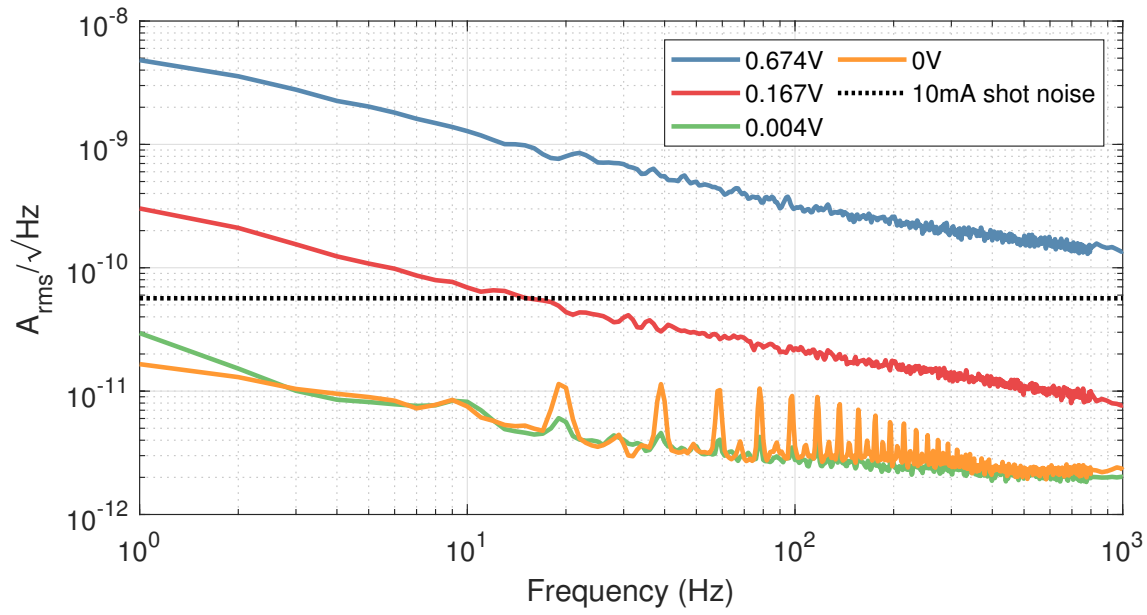


Figure A.2: IG26

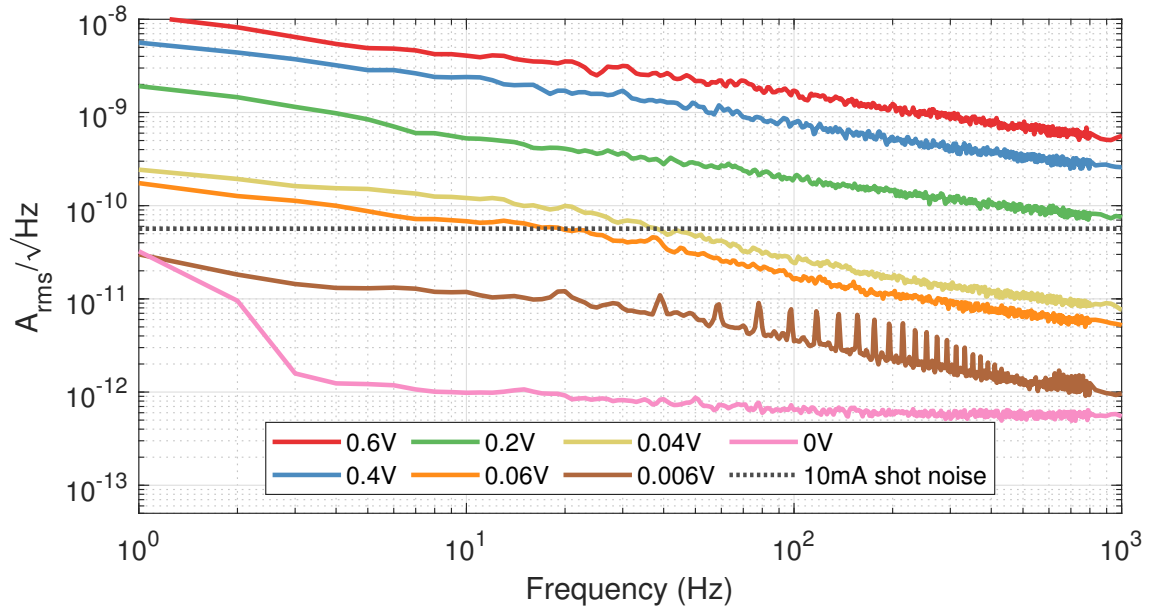


Figure A.3: IG22

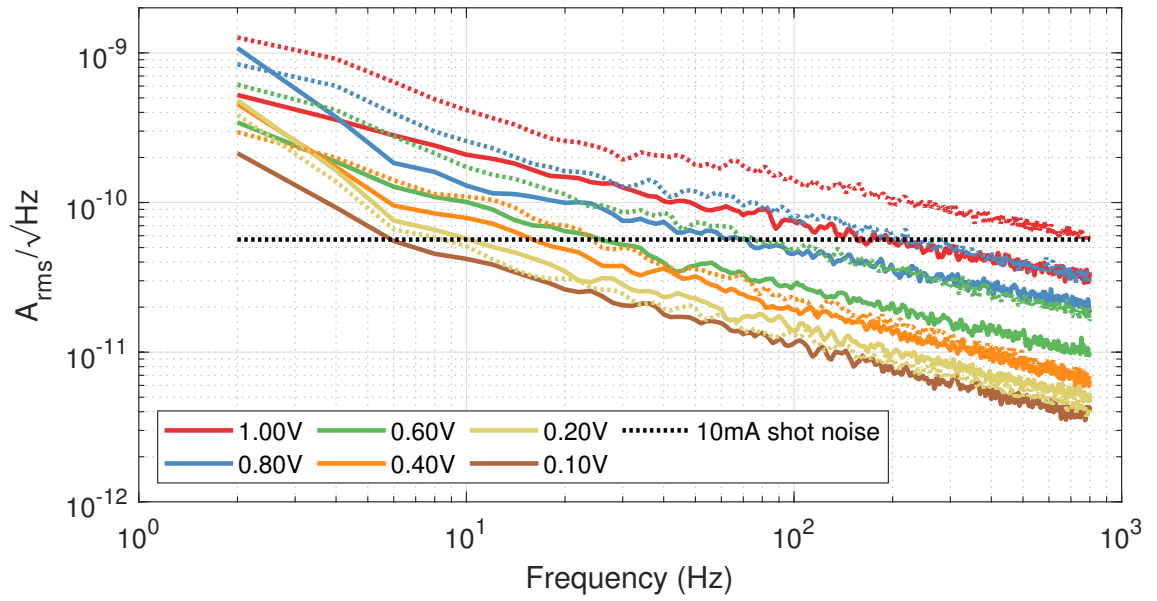


Figure A.4: G12183-010k

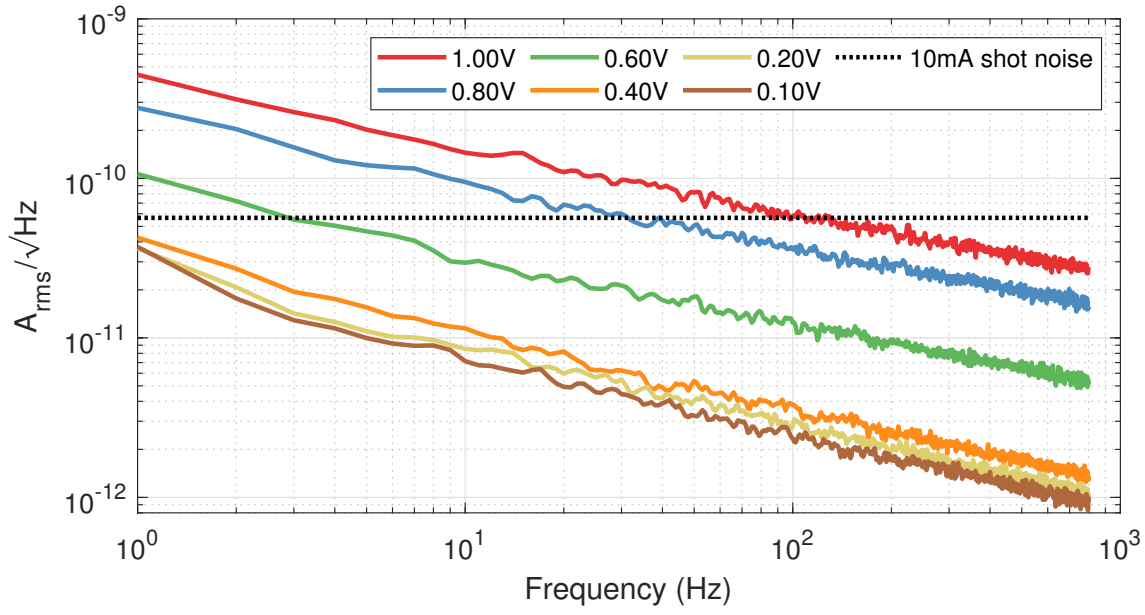


Figure A.5: G12183-005k

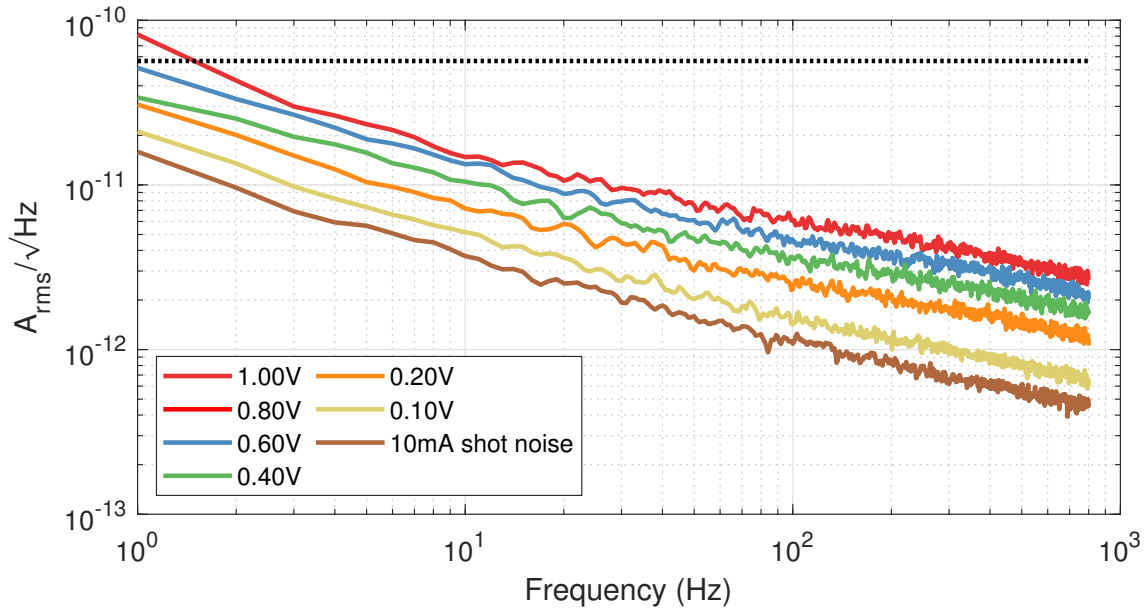


Figure A.6: G12182-010k

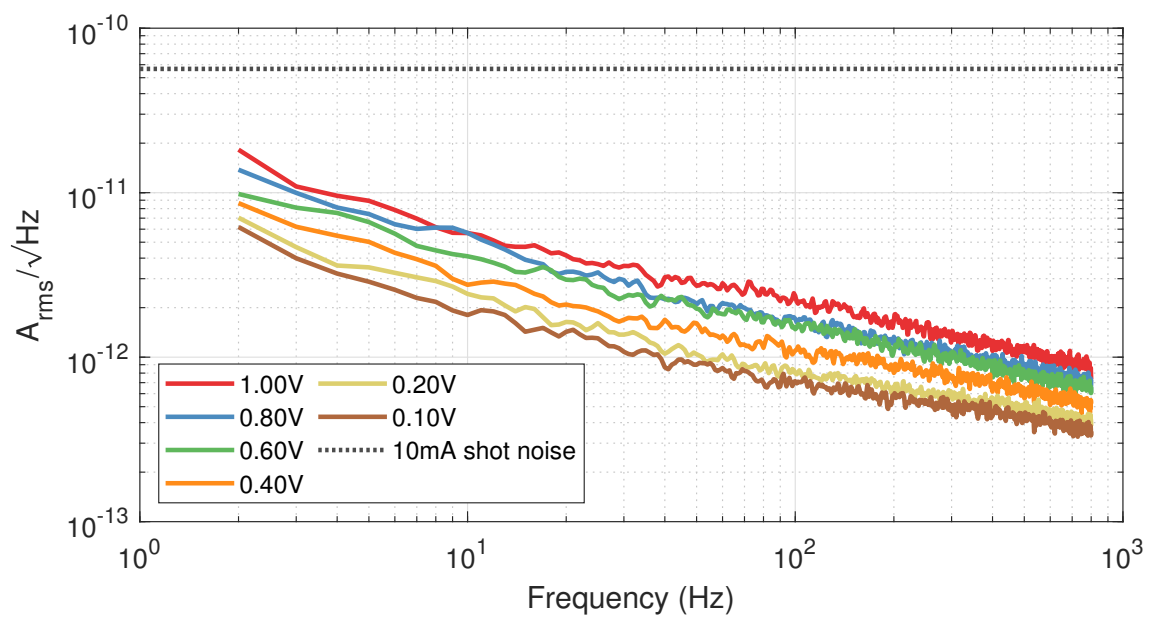


Figure A.7: G12182-005k

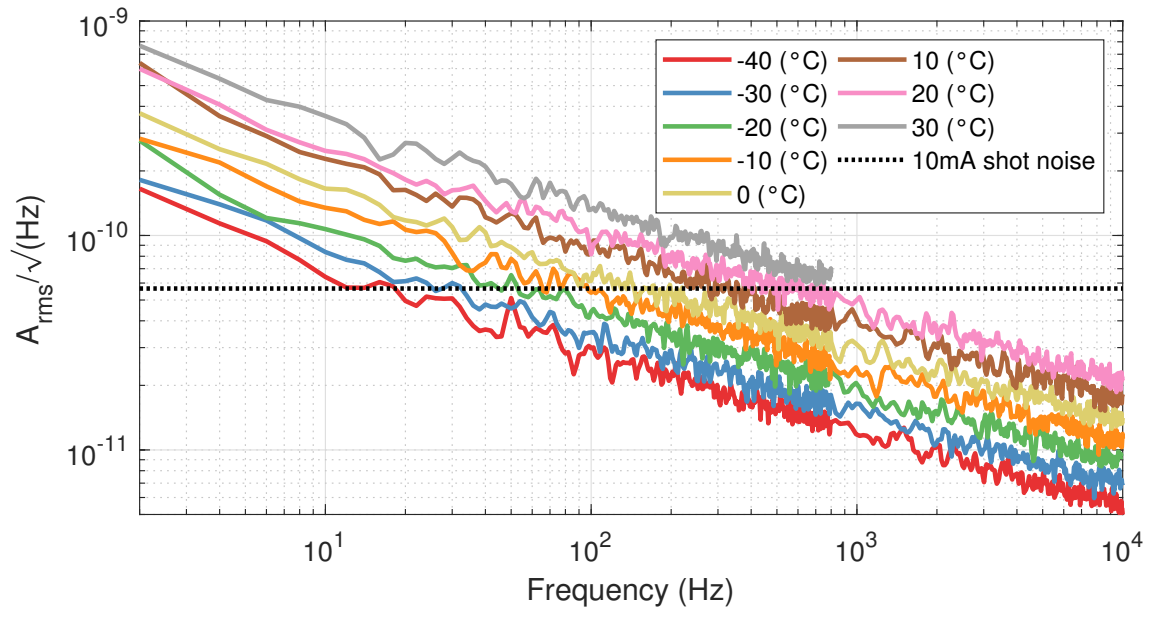


Figure A.8: FD10D 1.8V reverse bias

A.2 Temperature Section

A.3 InSb

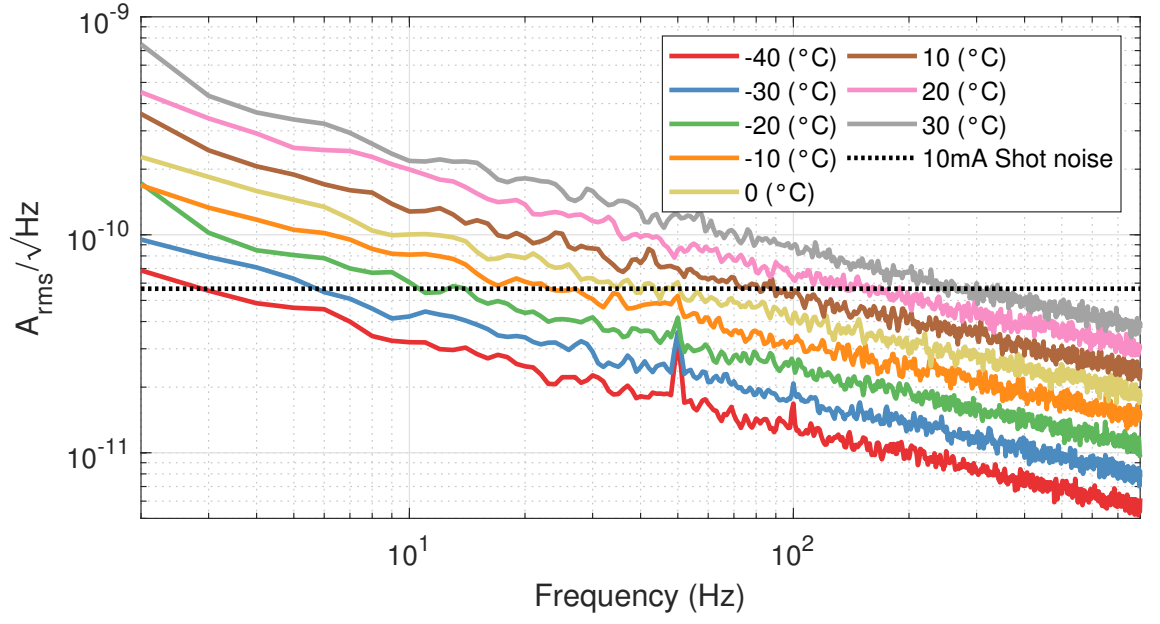


Figure A.9: G12183-010k 1.0V reverse bias

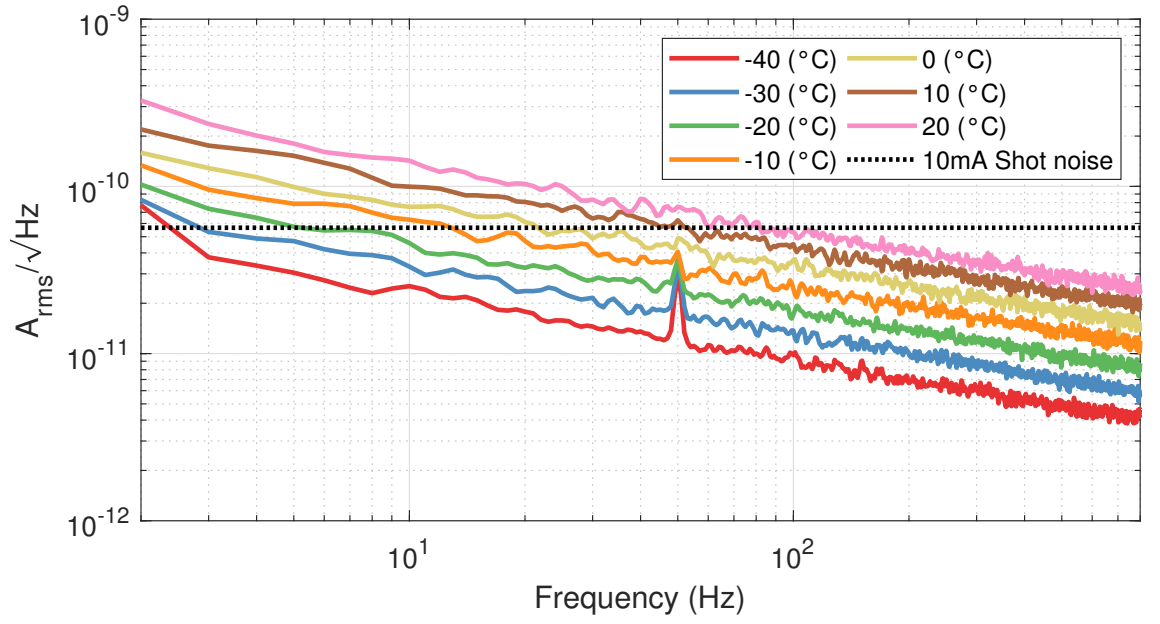


Figure A.10: G12183-005k 1.0V reverse bias

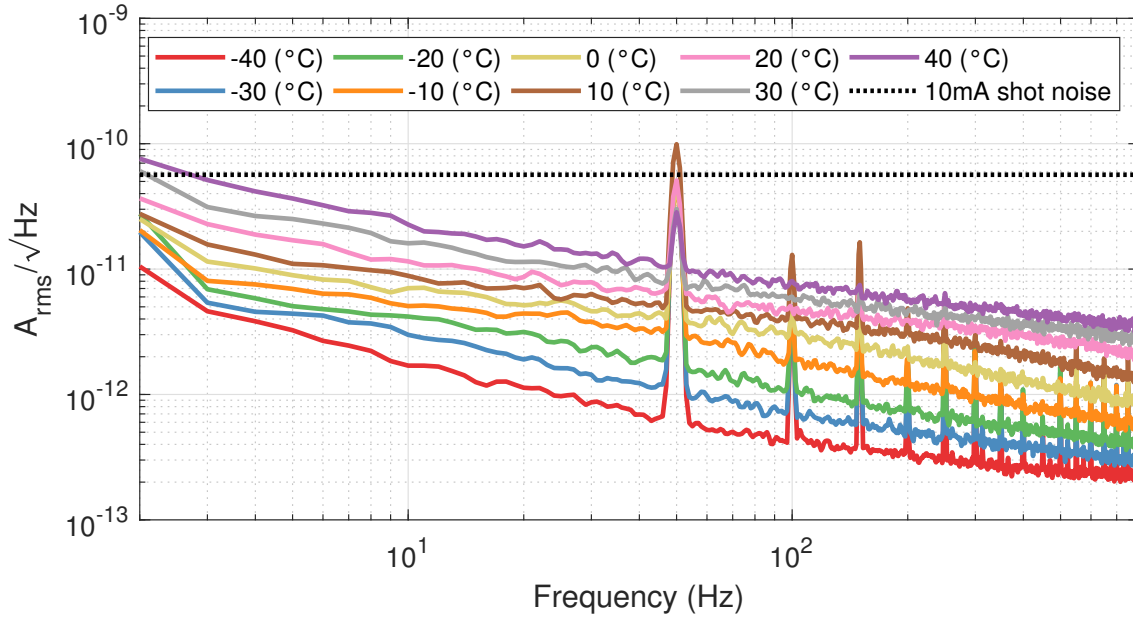


Figure A.11: G12182-010k 1.0V reverse bias

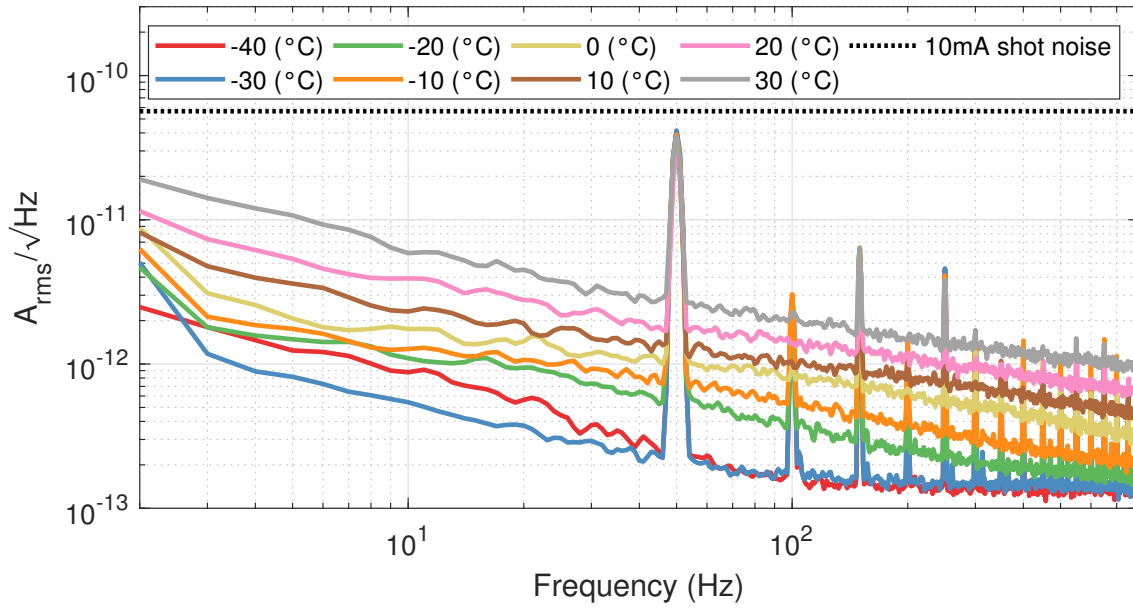


Figure A.12: G12182-005k 1.0V reverse bias

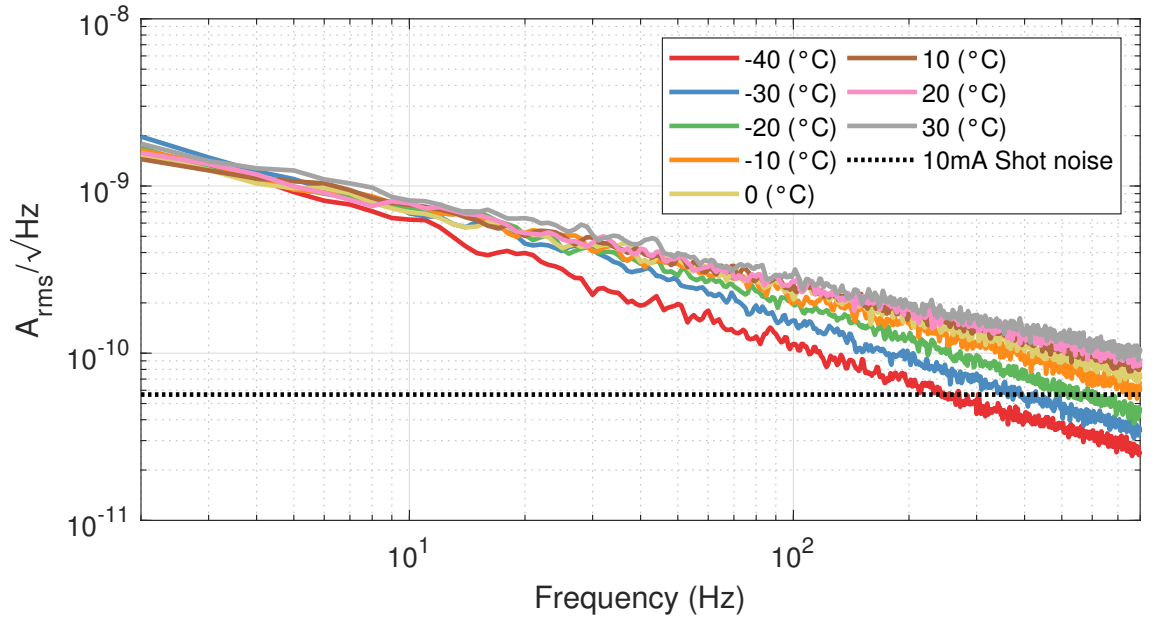


Figure A.13: IG24x500S4i 1.0V reverse bias

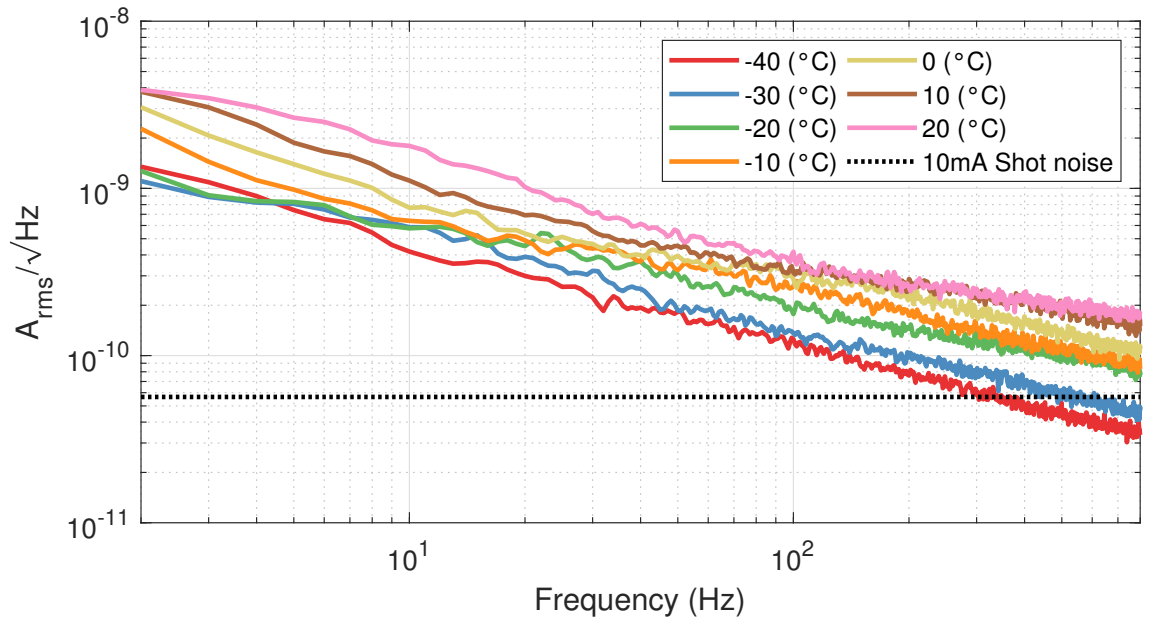


Figure A.14: IG26x500S4i 1.0V reverse bias

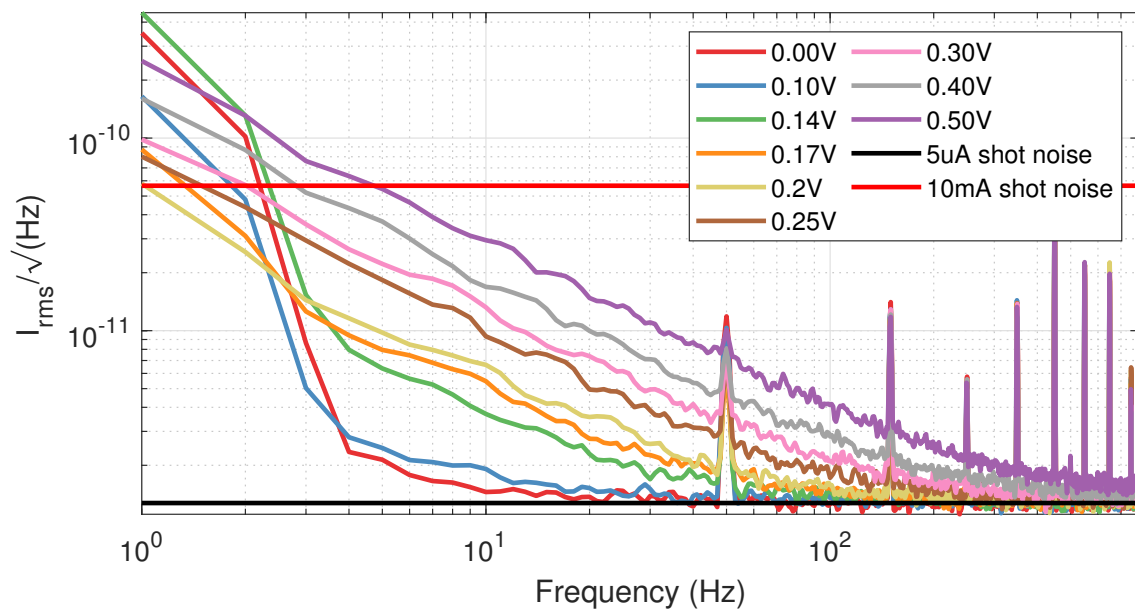


Figure A.15

Appendix B

Photodiode Circuits

B.1 Transimpedance Amplifier for Measuring the Dark Noise of Photodiodes

A circuit which allowed for the reverse bias to be set and for the easy exchange of photodiodes was constructed. To measure small amounts of current noise, a transimpedance amplifier with a large gain is needed. As the dark current for these diodes is of the order of 1mA, the gain of the transimpedance amplifier was chosen to be 1×10^6 so that signals of the order 1 V would be produced, and so a $1 \text{ M}\Omega$ resistor was used as in the feedback path.

Operation amplifiers have current noise at their inputs, voltage noise at the their outputs and the feedback resistor will generate Johnson-Nyquist noise (e.g. [20, Figure 8.58]). Designers of OP amps, generally, improve the OP amps output voltage noise at the expense of the input current noise or vice versa and so there is not one-size-fits-all op amp. In this application, as the current noise that is to be measured

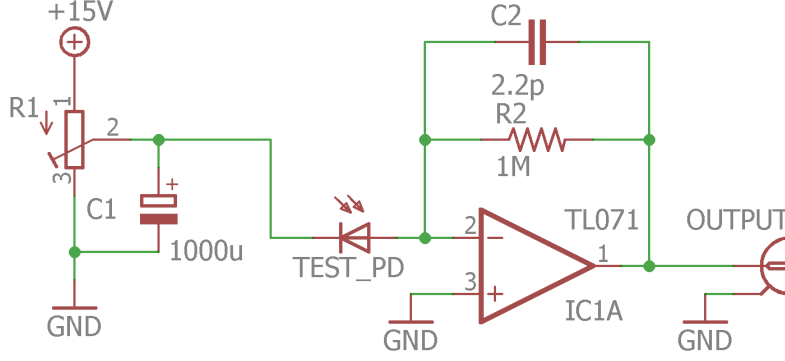


Figure B.1: Circuit used to measure the dark noise of photodiodes, represented by TEST_PD, under different reverse biases. The transimpedance amplifier has a $1\text{M}\Omega$ (R2) in parallel with a 2.2pF capacitor (C2) in the feedback path of the TL071 op amp (TL071P). The reverse bias was generated by a potentiometer (R1) connected to an LM7815 voltage regulator. This voltage was low passed using C1

is small and the feedback resistor is large, a TL071, a JFET style op amp with low input current noise, was selected. As the photodiode has capacitance and the op amp has finite bandwidth, a 2.2pF capacitor in parallel to keep the circuit stable.

Several methods of providing the reverse bias were tested. The first of these derived the bias from a LM7815 regulator in combination with a variable resistor configured a potential divider; this circuit is shown in Figure B.1. A similar circuit which was used in later experiments is shown in Figure B.2.

To verify that this was a sufficiently quiet bias voltage source that does not contaminate the measurements with noise, an ultra-quiet bias reference was used to cross check a measurement; this circuit is shown in Figure B.2. By filtering the voltage made a reference voltage IC (AD587JNZ), this circuit provided a bias with

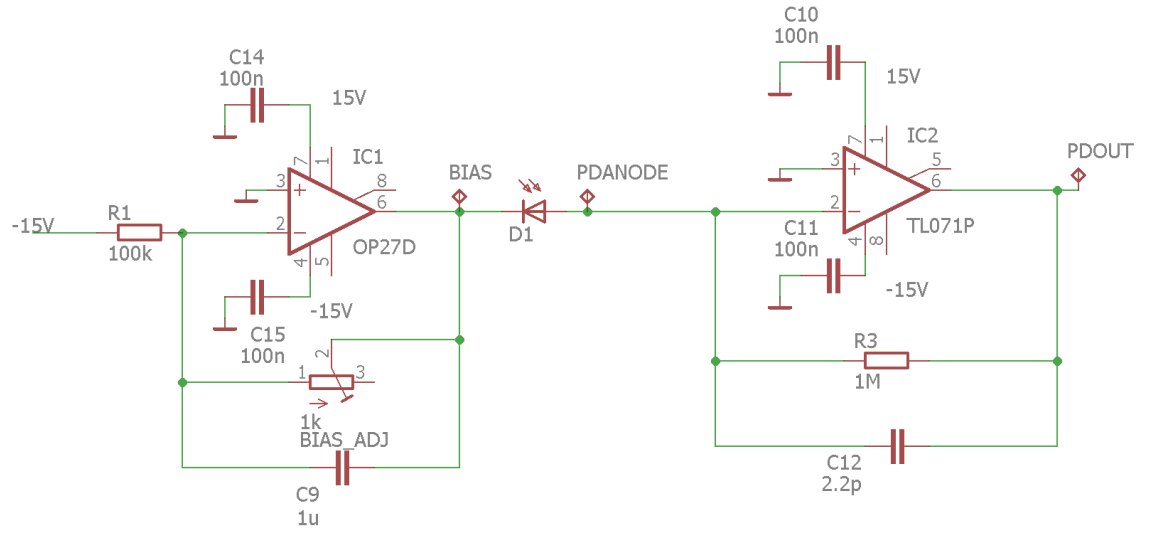


Figure B.2: Circuit used to measure the dark noise of photodiodes. The transimpedance amplifier used is identical to the one shown in Figure B.1. The bias voltage was generated using an OP27 op amp (OP27D) configured with so that the gain between the inverting input and the output of OP27D was > 1 .

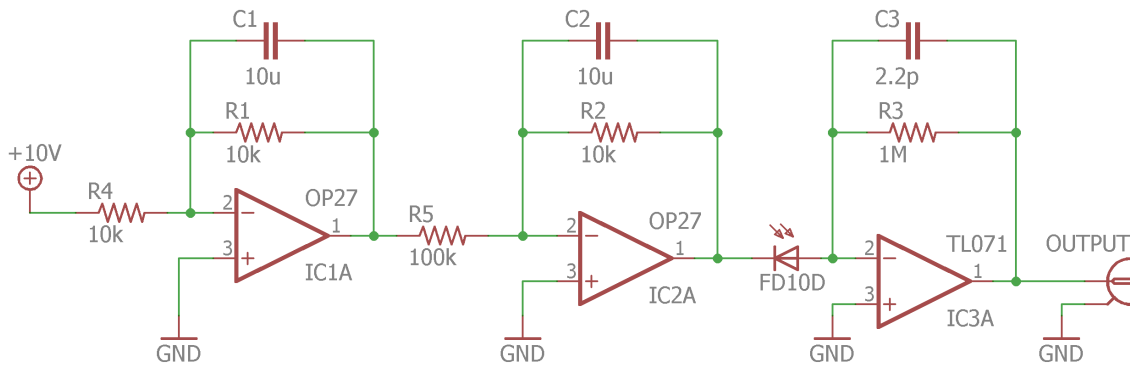


Figure B.3: Circuit used to check that bias voltage noise was not contaminating dark noise measurements. To reach $5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz, a 10V low noise reference IC (AD587JNZ) was used to generate the reverse bias. The output of this IC was filtered by two low pass filters (IC1A and IC2A). The transimpedance amplifier used is identical to the one shown in Figure B.1

$5\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz. The bias of the FD10D (see Section ??) was set to 1 V in both circuits. Figure B.4 shows that the regulator-potentiometer bias voltage source is sufficiently quiet.

B.1.1 Transfer function

The transfer function of the transimpedance amplifiers shown in Figure ?? was measured under a variety of different physical constructions. This circuit's behaviour can be affected by stray capacitance. When constructed from strip board and with the photodiode anode and cathode cables very near each other, there was of the order 10 pF of stray capacitance i.e. it was as if the circuit had an extra 10 pF capacitor between the inverting input of the op amp and ground. By making this circuit on a prototyping board with less stray capacitance between the strips and

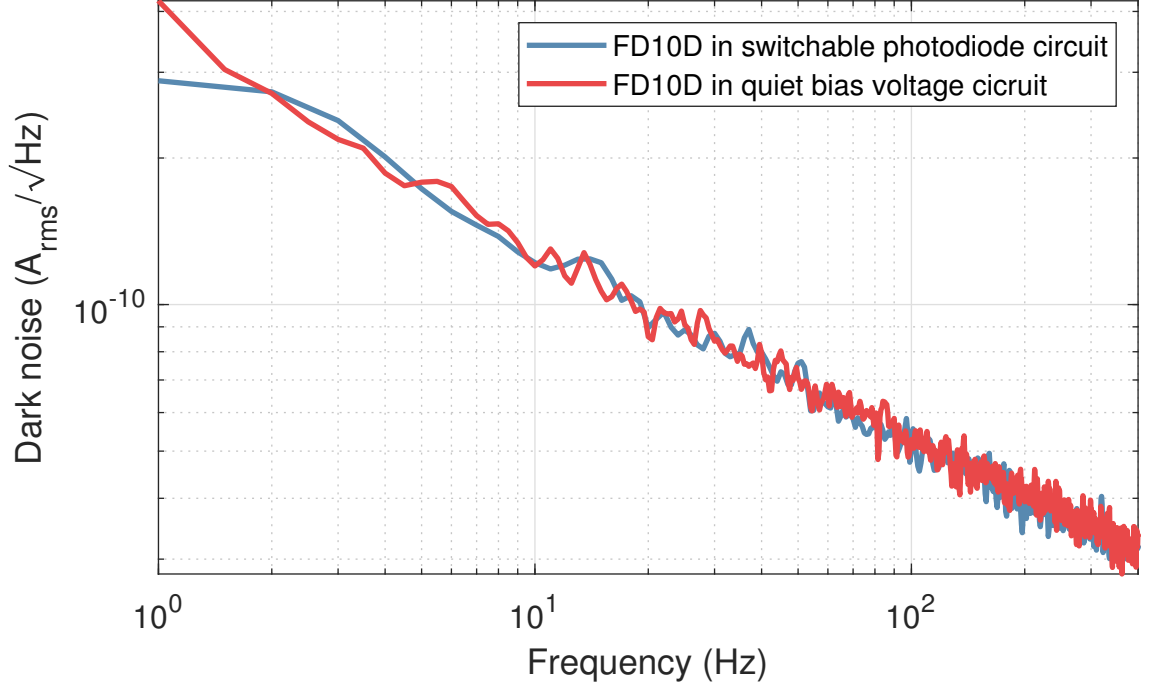


Figure B.4: The FD10D was biased with 1 V in the circuits shown in Figure B.1 and Figure B.2. The low noise fixed bias circuit and the circuit with a settable bias gave the same results, so we can be confident that the circuits shown in Figures B.1 and B.3 will produce measurements that are not contaminated by bias noise.

by keeping the photodiode anode and cathode wires from forming a capacitor, the stray capacitance was reduced. The measurement is in agreement with the LISO simulation of this circuit. This was cross checked using the method described in Section B.4.

B.1.2 Noise measurement

The dark noise of the circuits described in Section B.1 is shown in Figure B.6. The shot noise level of a typical photocurrent in a gravitational wave detector is several hundred times larger than the noise of this circuit. To measure noise of the transimpedance amplifier, instead of a photodiode, a 470 pF capacitor to was connected between the inverting input of the op amp and ground.

B.2 Transimpedance Amplifier for Making Shot Noise limited Measurements of 10 mA Photocurrents

A photodiode circuit which is shot noise limited over a up to several 100 kHz is shown in Figure B.7. A DC bias was created by low-passed bench-top lab power supply. An AC component could be added onto the bias via a summing OP-amp. The FD10D was connected to a transimpedance amplifier that whose electronic noise was below the shot noise level for ~ 10 mA currents. The noise of this circuit is shown in ... the transfer function of this circuit is shown in...

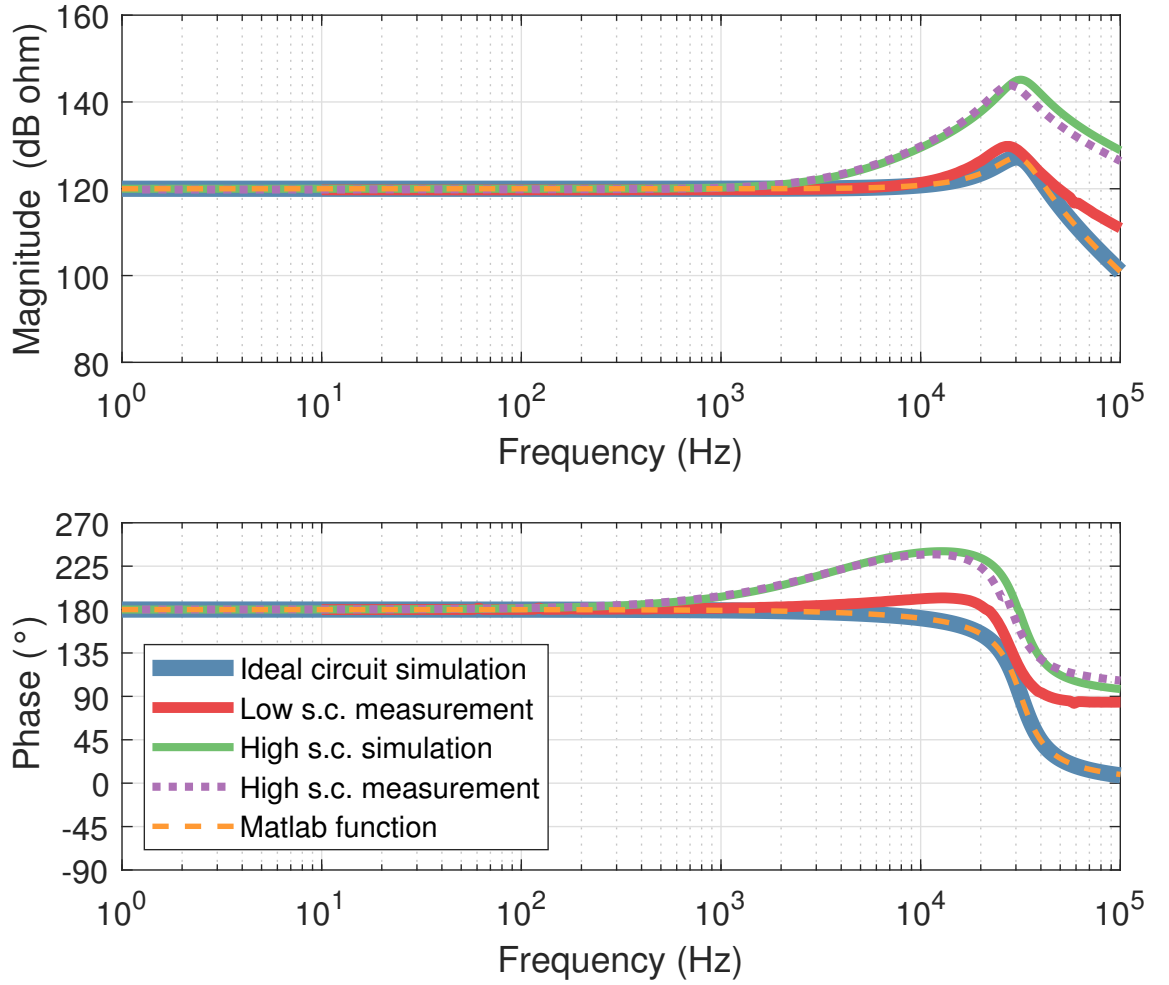


Figure B.5: The transfer function between a current into the inverting input of the transimpedance amplifiers designs shown in Figures B.1B.3 and B.2 was measured and simulated using LISO. A MATLAB function (see Section B.4) was used to check the measurement and the simulation. This circuit is sensitive to stray capacitance (s.c.) at the inverting input of the TL071. The stray capacitance was reduced by changing the layout of the circuit.

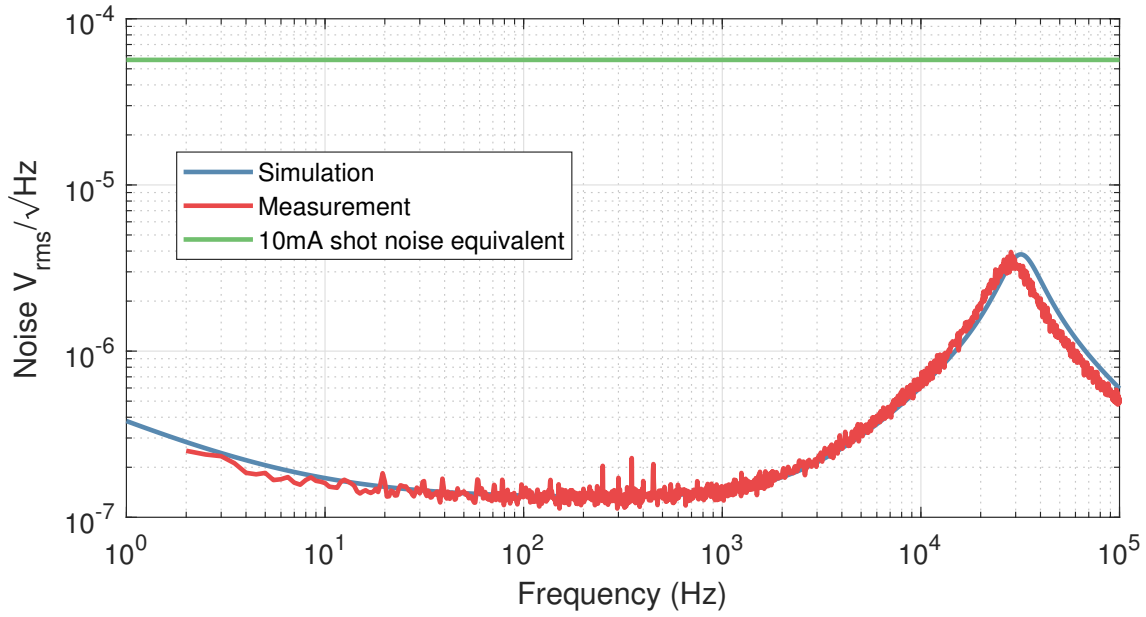


Figure B.6: At low frequencies, the noise of the circuit design shown in Figures B.1B.3 and B.2 is dominated by the low frequency voltage noise of the TL071. Above 10 Hz, the noise of the circuit is dominated by the johnson-Nyquist noise of the feedback resistor. At 30 kHz, the impedance of the feedback network causes the TL071's voltage noise to rise above the resistors noise.

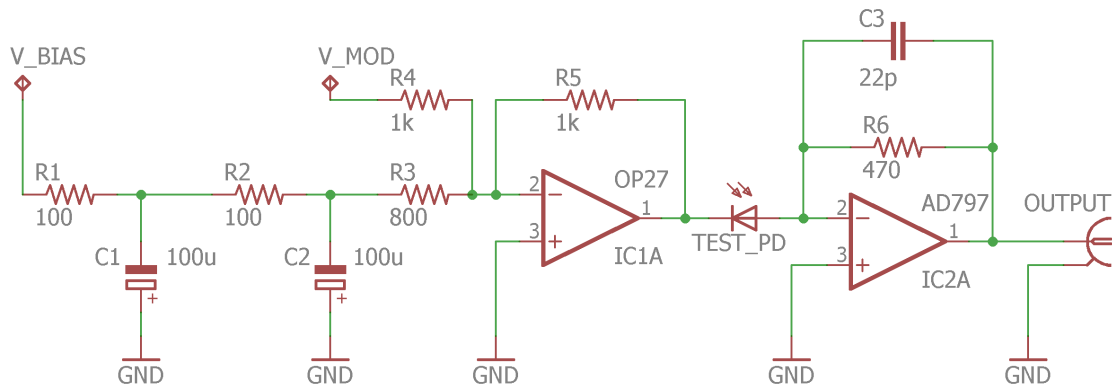


Figure B.7

B.3 Simulations with LISO

The behaviour of circuits was simulated using LISO. This software performs a nodal analysis to calculate the voltages at each point within a circuit. LISO can calculate the noise and transfer function of circuits that include op amps. LISO has a wide variety of op amps in its op amp library and this can be easily expanded by specifying the noise properties, open loop gain and gain-bandwidth frequency.

B.4 Transfer Function Calculation With MATLAB

The open loop transfer function of an op amp is

$$A(s) = \frac{A_0}{\frac{A_0}{2\pi f_{\text{GWB}}}s + 1}. \quad (\text{B.1})$$

The transfer function between the input current to the inverting input and the voltage at the output pin for an op amp made into a transimpedance amplifier is

$$H(s) = Z_{\text{FB}} \frac{A(s)}{1 + A(s)B(s)}, \quad (\text{B.2})$$

where $B(s)$ is the transfer function of the components between the output pin of the op amp and the inverting input of the op amp and Z_{FB} is the impedance seen by the current flowing from the output pin to the inverting input. Without a capacitor in parallel with the feedback resistor $B(s) = \frac{1}{R_{\text{FB}}C_{\text{PD}}s+1}$. This can be problematic as the transfer function of the circuit may have a phase of 180° due to the photodiode's capacitance and the gain bandwidth product of the op amp as it crosses the unity gain point which will cause the circuit to be unstable. To fix this problem, a capacitor

should be added in parallel to the feedback resistor, and so

$$B(s) = \frac{R_{\text{FB}}C_{\text{FB}}s + 1}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.3})$$

The impedance seen by the current is

$$Z_{\text{FB}} = \frac{R_{\text{FB}}}{R_{\text{FB}}(C_{\text{FB}} + C_{\text{PD}})s + 1}. \quad (\text{B.4})$$

$H(s)$ can be calculated with the following MATLAB code:

```
function [mag,phase] = photodiodeTransimpedance(cf, rf, cpd, gbw, A, w)
% cf is the capacitor in the feedback loop used to keep the circuit
% stable, and its units are Farads. rf is the feedback resistor
% measured in Ohms. cpd is the photodiodes capacitance (Farads).
    gbw is the
% gain bandwidth of the OP amp, and is measured in Hz. A is the
    Open loop
% gain of the OP amp. w is the angular frequency range that
% the phase and magnitude will be calculated over.
A_s = tf(A, [A/(2*pi*gbw), 1]);
B_s = tf([cf*rf, 1], [rf*(cf+cpd), 1]);
closedLoopAnalytic = -tf(rf, [rf*(cf+cpd), 1])*A_s/(1+A_s*B_s);
[mag, phase] = bode(closedLoopAnalytic, w);
mag = squeeze(mag);
phase = squeeze(phase);
end
```

Bibliography

- [1] B. Lantz, S. L. Danilishin, S. Hild, Gustafson E., D. Coyne, V. Quetschke, G. D. Hammond, R. Adhikari, M. Evans, and R. Bassiri. Instrument science white paper 2018. LIGO DCC.

- [2] M Punturo, M Abernathy, F Acernese, B Allen, N Andersson, K Arun, F Barone, B Barr, M Barsuglia, M Beker, N Beveridge, S Birindelli, S Bose, L Bosi, S Braccini, C Bradaschia, T Bulik, E Calloni, G Cella, E Chassande Mottin, S Chelkowski, A Chincarini, J Clark, E Coccia, C Colacino, J Colas, A Cumming, L Cunningham, E Cuoco, S Danilishin, K Danzmann, G De Luca, R De Salvo, T Dent, R De Rosa, L Di Fiore, A Di Virgilio, M Doets, V Fafone, P Falferi, R Flaminio, J Franc, F Frasconi, A Freise, P Fulda, J Gair, G Gemme, A Gennai, A Giazotto, K Glampedakis, M Granata, H Grote, G Guidi, G Hammond, M Hannam, J Harms, D Heinert, M Hendry, I Heng, E Hennes, S Hild, J Hough, S Husa, S Huttner, G Jones, F Khalili, K Kokeyama, K Kokkotas, B Krishnan, M Lorenzini, H Lück, E Majorana, I Mandel, V Mandic, I Martin, C Michel, Y Minenkov, N Morgado, S Mosca, B Mours, H Müller-Ebhardt, P Murray, R Nawrodt, J Nelson, R Oshaughnessy, C D Ott, C Palomba, A Paoli, G Parguez, A Pasqualetti, R Passaquieti, D Passuello, L Pinard,

- R Poggiani, P Popolizio, M Prato, P Puppo, D Rabeling, P Rapagnani, J Read, T Regimbau, H Rehbein, S Reid, L Rezzolla, F Ricci, F Richard, A Rocchi, S Rowan, A Rüdiger, B Sassolas, B Sathyaprakash, R Schnabel, C Schwarz, P Seidel, A Sintès, K Somiya, F Speirits, K Strain, S Strigin, P Sutton, S Tarabrin, A Thüring, J van den Brand, C van Leewen, M van Veggel, C van den Broeck, A Vecchio, J Veitch, F Vetrano, A Vicere, S Vyatchanin, B Willke, G Woan, P Wolfango, and K Yamamoto. The einstein telescope: a third-generation gravitational wave observatory. *Classical and Quantum Gravity*, 27(19):194002, sep 2010.
- [3] The LIGO Virgo Collaboration. Exploring the sensitivity of next generation gravitational wave detectors. *Classical and Quantum Gravity*, 34(4):044001, jan 2017.
- [4] Georgia L. Mansell, Terry G. McRae, Paul A. Altin, Min Jet Yap, Robert L. Ward, Bram J. J. Slagmolen, Daniel A. Shaddock, and David E. McClelland. Observation of squeezed light in the $2\mu\text{m}$ region. *Physical Review Letters*, 120(20), may 2018.
- [5] M. J. Yap, D. W. Gould, T. G. McRae, P. A. Altin, N. Kijbunchoo, G. L. Mansell, R. L. Ward, D. A. Shaddock, B. J. J. Slagmolen, and D. E. McClelland. Squeezed vacuum phase control at $2\mu\text{m}$. *Optics Letters*, 44(21):5386, nov 2019.
- [6] Alan R. Beattie. Quantum efficiency in InSb. *Journal of Physics and Chemistry of Solids*, 23(8):1049–1056, aug 1962.

- [7] S. Forrest. Performance of $\text{In}_x\text{Ga}_{1-x}\text{As}_y\text{P}_{1-y}$ photodiodes with dark current limited by diffusion, generation recombination, and tunneling. *IEEE Journal of Quantum Electronics*, 17(2):217–226, feb 1981.
- [8] F N Hooge, T G M Kleinpenning, and L K J Vandamme. Experimental studies on $1/f$ noise. *Reports on Progress in Physics*, 44(5):479–532, may 1981.
- [9] M. Gendry, V. Drouot, C. Santinelli, and G. Hollinger. Critical thicknesses of highly strained InGaAs layers grown on InP by molecular beam epitaxy. *Applied Physics Letters*, 60(18):2249–2251, may 1992.
- [10] Ramon U. Martinelli, Thomas J. Zamerowski, and Paul A. Longeway. $2.6\ \mu\text{m}$ InGaAs photodiodes. *Applied Physics Letters*, 53(11):989–991, sep 1988.
- [11] Lars Zimmermann, Joachim John, Stefan Degroote, Gustaaf Borghs, Chris Van Hoof, and Stefan Nemeth. Extended wavelength InGaAs on GaAs using InAlAs buffer for back-side-illuminated short-wave infrared detectors. *Applied Physics Letters*, 82(17):2838–2840, apr 2003.
- [12] Gregory M Harry and the LIGO Scientific Collaboration. Advanced LIGO: the next generation of gravitational wave detectors. *Classical and Quantum Gravity*, 27(8):084006, apr 2010.
- [13] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [14] Teng Zhang, Denis Martynov, Andreas Freise, and Haixing Miao. Quantum squeezing schemes for heterodyne readout. *Physical Review D*, 101(12), jun 2020.

- [15] M Prijatelj, J Degallaix, H Grote, J Leong, C Affeldt, S Hild, H Lück, J Slutsky, H Wittel, K Strain, and K Danzmann. The output mode cleaner of GEO 600. *Classical and Quantum Gravity*, 29(5):055009, feb 2012.
- [16] Evan D. Hall, Kevin Kuns, Joshua R. Smith, Yuntao Bai, Christopher Wipf, Sebastien Biscans, Rana X Adhikari, Koji Arai, Stefan Ballmer, Lisa Barsotti, Yanbei Chen, Matthew Evans, Peter Fritschel, Jan Harms, Brittany Kamai, Jameson Graef Rollins, David Shoemaker, Bram Slagmolen, Rainer Weiss, and Hiro Yamamoto. Gravitational-wave physics with cosmic explorer: limits to low-frequency sensitivity.
- [17] M. López, J. C. Molina, H. Hofer, A. Sperling, and S. Kück. Measurement of the nonlinearity of ge-and InGaAs-photodiodes at high irradiance levels. *MAPAN*, 25(1):47–52, mar 2010.
- [18] Ibrahim Kimukin, Necmi Biyikli, and Ekmel Ozbay. InSb high-speed photodetectors grown on GaAs substrate. *Journal of Applied Physics*, 94(8):5414, 2003.
- [19] A.J. Moseley, M.D. Scott, A.H. Moore, and R.H. Wallis. High-efficiency, low-leakage MOCVD-grown GaInAs/AlInAs heterojunction photodiodes for detection to $2.4\mu\text{m}$. *Electronics Letters*, 22(22):1206, 1986.
- [20] P. Horowitz and W. Hill. *The Art of Electronics*. Cambridge University Pr., 2015.